Team Sammard: Project AGNEYA

Team 145 Project Technical Report to the 2024 Spaceport America Cup

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VIT Vellore, Team Sammard 145

The following document describes the design and test specifications of Team Sammard's rocket- 'AGNEYA' for IREC 2024. The rocket is designed to achieve an apogee of 10,000 ft AGL. The rocket is our entry into Spaceport America Cup 2024 for the 10k COTS category. We have an 8.8-pound experimental payload. The Active Control Gimbal Stabilization payload integrates advanced Gimbal BLDC motors and Magnetic Encoders to create a stable platform. The nose cone is optimized for low drag using the LD Haack profile. An M class solid rocket motor was used to power the rocket (Aerotech M1845NT-PS). The motor delivers a peak thrust of 2434 N and impulse of 8093 Ns. The airframes are made out of fiberglass composite. The nose cone is made out of carbon fiber with an aluminum tip. Internal bulkheads and the fins are made out of fiberglass. The avionics of the rocket feature Missileworks RRC3 COTS (primary) and Featherweight Raven (redundant). Our very own SRAD Flight computer SPARC is also part of the avionics bay. We use Black powder based recovery mechanism that has been rigorously tested with the drogue parachute deploying at the apogee and the main parachute to be deployed at around 1500 ft AGL. Trajectory simulations have been done using the latest Open Rocket software.

Nomenclature:

COTS =	Commercial off the shelf
CP =	Centre of Pressure
Cg =	Center of gravity
c =	Chord
dt =	Time step
Fx =	X component of the resultant pressure force acting on the vehicle
Fy =	Y component of the resultant pressure force acting on the vehicle
f, g =	Generic functions
h =	Height
i =	Time index during navigation
j =	Waypoint index

I. INTRODUCTION

Team Sammard, a vibrant student rocketry team nestled within VIT University in Vellore, India, shines brightly as one of the institute's most esteemed organizations. With steadfast academic support and access to dedicated workspaces, this team flourishes under the nurturing guidance of their beloved university. Financially, Team Sammard stands on solid ground, with substantial backing from the university forming a crucial pillar of support. However, the true strength of their financial foundation lies in the unwavering commitment of their members and esteemed alumni, who generously contribute to the team's endeavors. At its core, Team Sammard is a dynamic learning laboratory, providing invaluable hands-on experience to its members. Through a dedication to fostering a collaborative and inter-departmental work environment, the team offers a unique advantage that cultivates individual growth. Despite occasional resource constraints, particularly in challenging times, Team Sammard remains steadfast in its commitment to community outreach. Serving as a beacon of enlightenment, the team hosts a variety of engaging webinars, courses, and interactive sessions, many of which have become cherished traditions.

1. ACADEMIC PROGRAM

Team Sammard, a distinguished sounding rocket team based at Vellore Institute of Technology (VIT), India, benefits from a range of university amenities, including access to funding requests, financial system management, and facilities. While the university provides essential funding, the team also actively raises additional resources through donations. Composed primarily of engineering students, Team Sammard utilizes Project Labs' state-of-the-art manufacturing and testing facilities to explore and refine rocket concepts. Embracing a multidisciplinary approach, the team actively recruits members from diverse academic backgrounds, mirroring the collaborative environment found in industry settings. This diversity fosters the development of complex systems and encourages innovative problem-solving. The team's overarching goal is to bridge academic knowledge with real-world applications, offering members invaluable experience relevant to future careers in the industry. Furthermore, Team Sammard is committed to nurturing the talents of younger students, providing them with early exposure to technical and communication skills essential for success in their academic journeys.

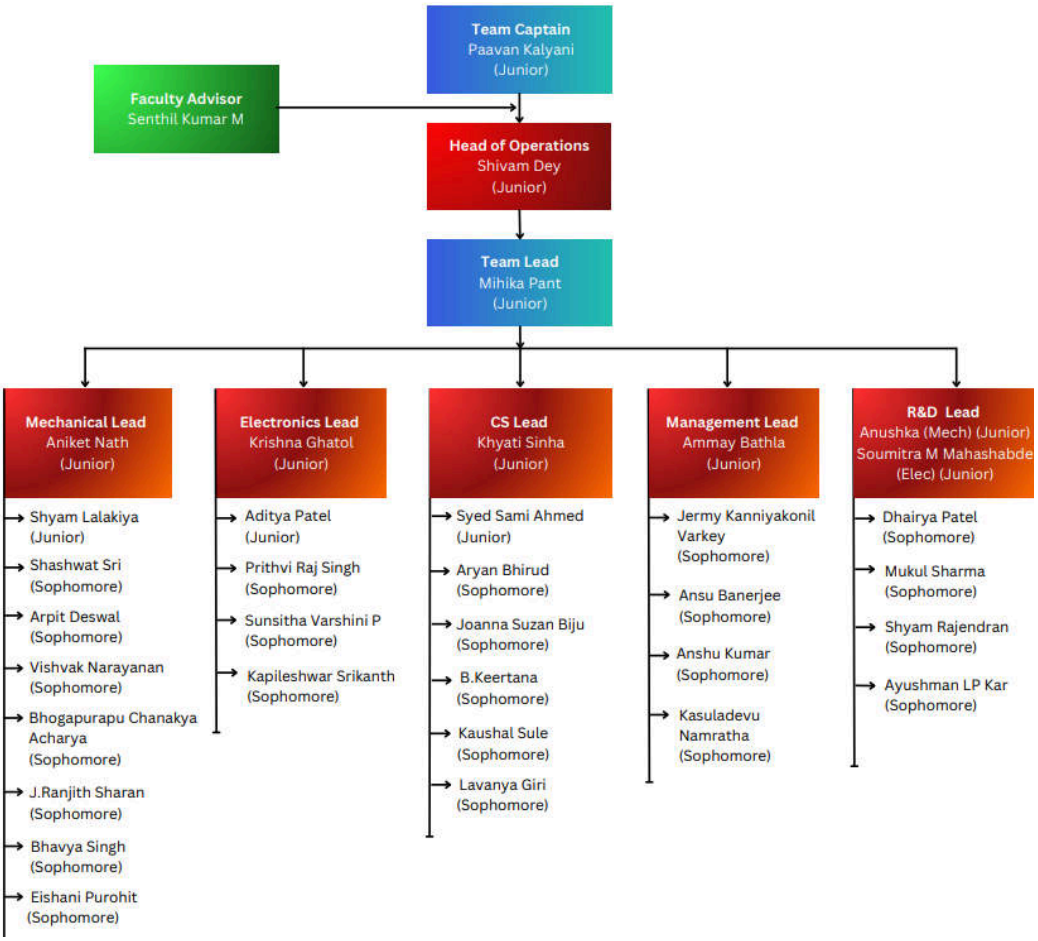
2. STAKEHOLDERS

As a relatively young team in the field of rocketry, Team Sammard relies primarily on funding from Vellore Institute of Technology, Vellore, to support its endeavors. The commitment and passion of its members often extend beyond graduation, with many continuing to support the team long after their time at the university. In addition to university funding, the team benefits from financial contributions from individuals and dedicated alumni who are invested in the team's success. Moreover, team members themselves contribute financially to sustain the operations of the team, demonstrating their dedication and commitment to its mission. The invaluable support of the university and monetary providers has been instrumental in the team's development journey. Their contributions serve as the foundation for current projects and pave the way for future endeavors. Furthermore, alumni play a crucial role by offering technical expertise, solutions, and guidance, serving as mentors to current team members as they navigate challenges and pursue innovation in rocket design and manufacturing.

3. TEAM MANAGEMENT STRATEGIES

Sammard's management strategy is anchored in three core principles: accountability, punctuality, and technical competence. Open communication is paramount, facilitated through digital, oral, and written channels. Virtual communication thrives on WhatsApp Messenger, where each department and sub-department manages dedicated groups for discussions and inquiries. At the onset of each project season, senior management and department heads collaborate to outline a comprehensive work timeline and define both long-term and short-term goals. Daily meetings at 9:30 PM in the university's Project Labs foster collaboration across departments, with weekly reviews to ensure alignment with objectives. To foster skill development and knowledge transfer, regular training sessions, technical presentations, and individual/group tasks are conducted by sub-department leads. Department heads bear the responsibility of ensuring technical proficiency while meeting project timelines. Monthly team outings foster cohesion, while comprehensive documentation—stored on Google Drive and GitHub—facilitates smooth operations and progress tracking. Weekly team and department meetings reinforce accountability and enable task allocation. Reflection is key at the end of each season, with senior management evaluating achievements and areas for improvement. Insights from these reviews inform adjustments to the team structure and project requirements, with leadership roles transitioning to younger members to sustain continuity and fresh perspectives.

4. TEAM STRUCTURE



II. SYSTEM ARCHITECTURE OVERVIEW

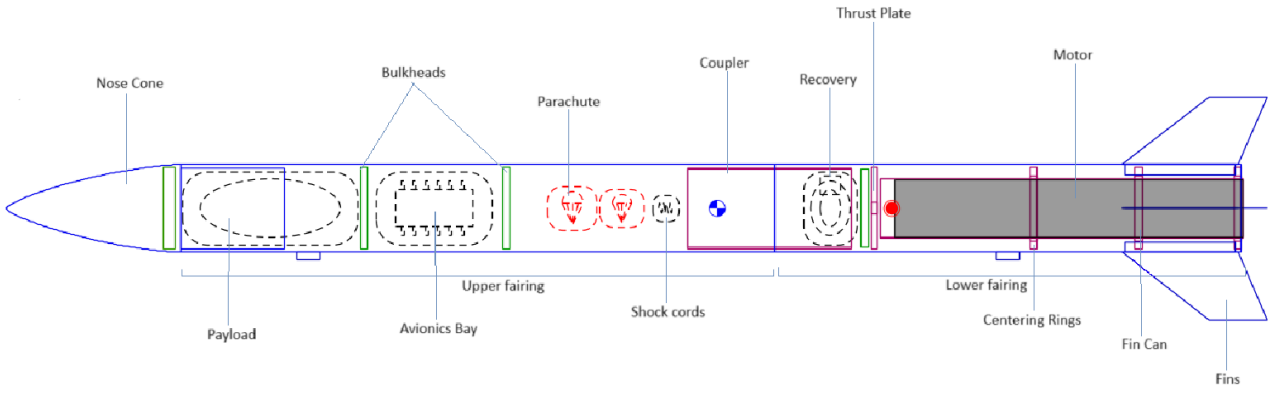


Fig: OpenRocket design

AGNEYA incorporates core engineering and design values like simplicity, workability, ease of manufacturing, robustness and experimentation. The team launched our first rocket in Spaceport America Cup 2023. Since then, we have grown a lot and learnt from our shortcomings to improve our design this time. In the following sections, the design decisions, rationale, analysis and manufacturing methods are discussed for the structure and avionics of the rocket.

A. COTS based Solid Propulsion System

The rocket uses a M class Aerotech motor of designation M1845NT-PS type to achieve a stable flight and desired apogee. With a total impulse of 8093 Ns and a peak thrust of 2434 N, it is an appropriate motor for a rocket our size. It has a burn time of 4.44 s. The high initial thrust of the motor helps in a stable ascent and a suitable off rail velocity. The details and thrust curve of the motor are shown below.

Table 1: Key features of M class motor and Thrust curve

Diameter	98 mm	Impulse	8093 Ns
Length	597 mm	Burn Time	4.44 s
Peak Thrust	2434 N	Average Thrust	1822 N
Total Weight	6682 g	Empty Mass	2910 g

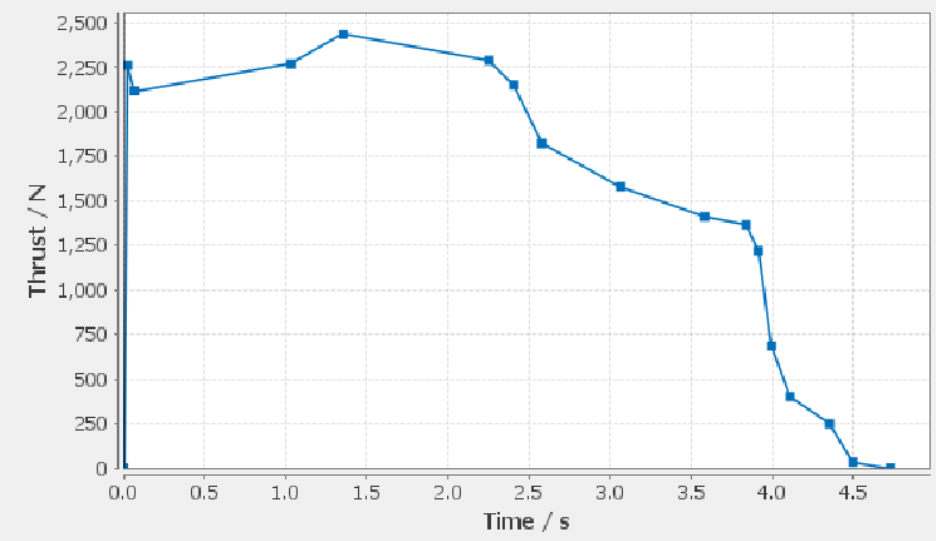


Fig: Thrust curve of M1845NT-PS

B. Flight Mechanics

The team made use of the updated version of the OpenRocket software (version 23.09) to design and conduct flight simulations. Any subsequent design changes were then again put through the simulations to find the design that is the most feasible and it is ensured that the flight is stable enough during the take off and throughout the flight.

The team used OpenRocket as the primary simulation software as it utilizes the extended Barrowman calculation method and a 6-DOF Runge-Kutta calculation method to precisely predict the flight profile and provide us with detailed flight data.

The first step was to try and simulate using the probable launch conditions during the time of the competition and to take into consideration the worst-case scenarios for better and more accurate results. Atmospheric conditions at the launch site were considered by observing its periodic trend at the time of launch, and taking into account the recommendations provided in the Design Test Evaluation Guide.

The launch rail that we use is provided by ESRA and is 17 ft long. Our simulations with the Average wind speed of 6.56 ft/s using standard temperature and pressure show a stable flight with suitable apogee. The flight profile graph is given in the figure below.

The rocket maintains a stability of about 2 to 3 Calibers throughout the flight and the velocity off launch rail is about 90 ft/s and any small variation in mass will not cause significant reduction. The stability doesn't exceed 3, preventing overstability.

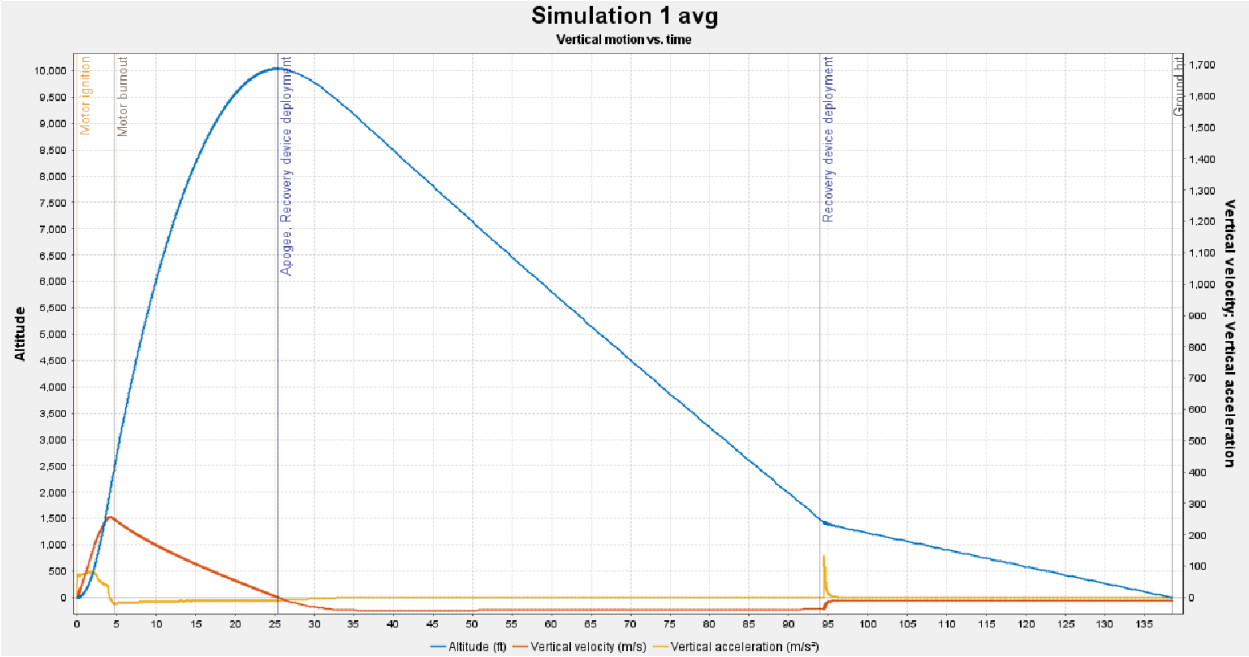


Fig: Flight profile graph

C. Aerostructures

The following is a detailed outline of the design, analysis and manufacturing of all the aerostructures. Softwares like Solidworks, Ansys Mechanical and Fluent were used in the process.

1. Nose Cone

DESIGN:
LD-Haack Series with shape parameter of 0.0 is used for the nose cone profile of Agneya. It is proved that this parameter develops least drags and achieves maximum apogee.

To determine the best possible design which produces least drag and maximum altitude, firstly different nose cone profiles were considered and analyzed. Agneya's flight regime is expected to lie in the subsonic region with a maximum Mach number of 0.8 at maxQ. This posed the challenge of selecting a nose cone profile which produces least drag at the maxQ. The team chose a profile with tangential transition to the body tube to prevent premature separation.

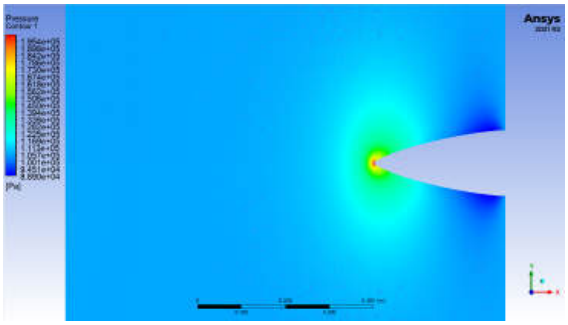
Three profiles were considered and analyzed, LD-Haack, Parabolic and Ellipse. OpenRocket is used to analyze these geometries by varying the profile and studying the behavior of Apogee attained and Drag. These results were used as bases for further analysis in which CFD simulations were done. These CFD simulations provided detailed analysis of pressure variation and drag properties for each of the profiles.

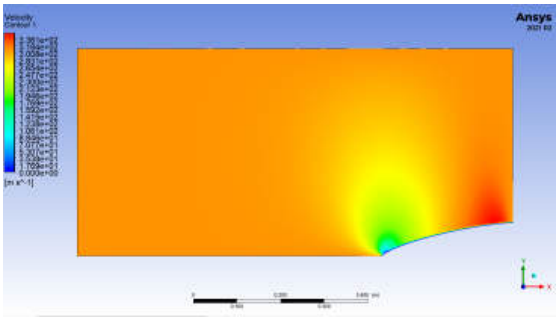
With the data obtained from OpenRocket, the case was simulated in CFD software. This was done to study the pressure distribution over nose cone profiles and to get a better understanding of drag characteristics. ANSYS Fluent was used for the CFD analysis.
The CFD analysis was done at the point of MaxQ, to obtain the worst possible pressure gradient and maximum drag generated at the maximum velocity.
A density-based setup was used for the boundary value problem of an axisymmetric model of nose cones. The gauge temperature and pressure at maxQ were calculated with the isentropic flow equation given by:

$$\frac{P}{Pt} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{-\gamma}{\gamma - 1}}$$

$$\frac{T}{Tt} = \left(\frac{P}{Pt}\right)^{\gamma/(\gamma-1)}$$

Where, Pt is the total pressure, P is the static pressure, Tt is the total temperature, T is the static temperature , γ is the ratio of specific heats and M is the Mach number at maxQ. Assuming γ = 1.4, The necessary calculations were done to set up the boundary value problem.





With the Reynold's number at the maxQ altitude, the appropriate wall spacing was calculated for the mesh. This was done to get an acceptable y+ value and accurate wall shear stress value.

The results produced by the LD-Haack profile suggest that the drag produced is the least and apogee attained is the highest in the case. The equation used for the construction of LD-Haack profile is as follows:

$$Y = \frac{R}{\sqrt{\pi}} * \sqrt{\theta - \frac{sin2\theta}{2}} \qquad \theta = \arccos\left(1 - \left(\frac{2x}{L}\right)\right)$$

Where,
R is the radius at the base of the nose cone
L is the length of the nose cone

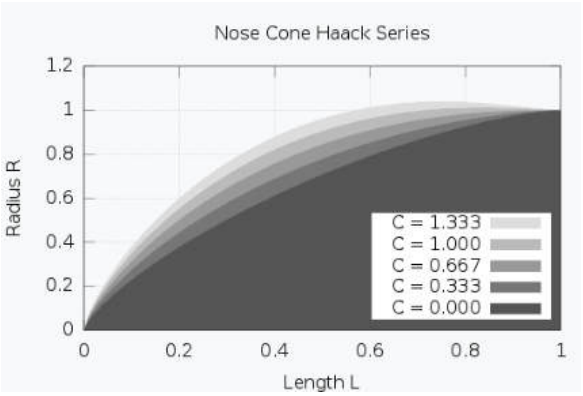


Fig: Haack Series profile

MANUFACTURING:
The LD Haack profile was chosen due to its aerodynamic superiority in the subsonic regime. The material chosen for the nose cone is Carbon Fiber due to its low weight compared to glass fiber or Aluminum. Wet layup process was used for the manufacturing of the nose cone. For this method, a mandrel is needed to be used as a die. A 3D printed mandrel is used for the same. Other substitutes like high density foam or MDF could have been used but were avoided to save cost and for ease of layup. The 3D printed mandrel is designed as the nose cone profile with proper tolerances. The size of the mandrel was determined considering the heating effect due to curing of the epoxy.
The material chosen is 200 GSM carbon fiber fabric. The material properties:

Properties	Units	Values
Tensile Strength	GPa	3.45
Tensile Modulus	GPa	230

2. Airframes

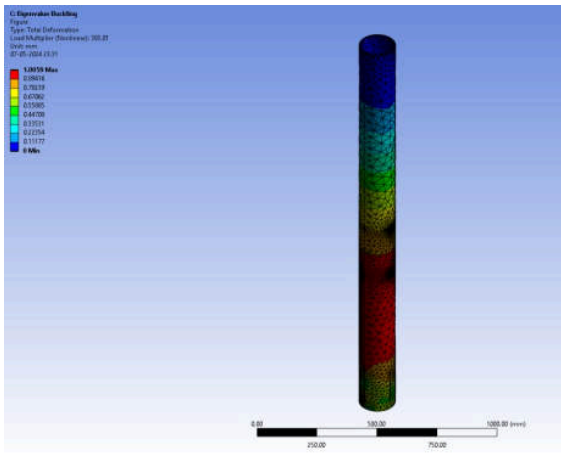
DESIGN:
The body tubes play a vital role as the primary load-bearing structures of the rocket, forming its outer casing. They endure substantial stress due to the integration of various components onto the airframe. The geometric requirements of the body tubes are influenced by the internal subsystems of the rocket. Specifically, the avionics bay is positioned directly above the recovery system within the upper body tube, facilitating easy access if necessary, during critical moments. The payload is situated between the upper body tube and the nose cone, allowing easy access. The inner diameter of the body tube is determined by factors such as the dimensions of the rocket motor and the available mould sizes. Overall, the rocket's external configuration consists of two body tubes.

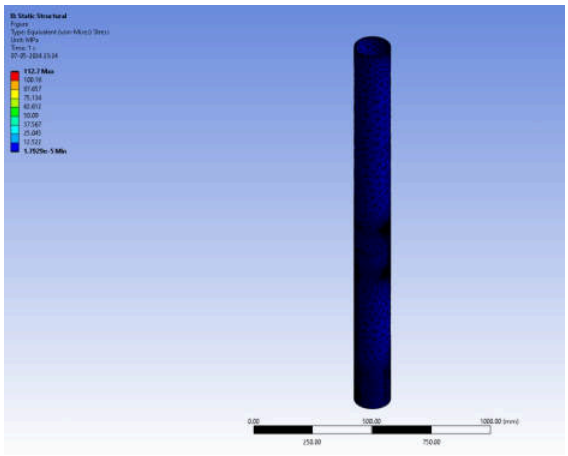
MANUFACTURING:



Fig: Body tubes and nose cone

The primary challenge anticipated for the body tubes is the risk of buckling, which necessitates their design to withstand the high impulse generated by the rocket motors. To validate the design, a static structural analysis was conducted using the ANSYS software. The composite geometry was constructed utilizing the ANSYS Mechanical module. Glass fiber woven material was employed for the analysis, possessing a tensile modulus of 3400MPa.
The thickness of the tube was chosen to be 5mm. A force of 10000N was applied for analysis purposes, which is higher than the actual forces experienced by the body tubes by 4-5 times. The simulation was to analyze if the part fails under the load from the motor or if it shears. Very low deformation values confirmed that the part will withstand the forces applied onto it. This validated the efficacy of the design.

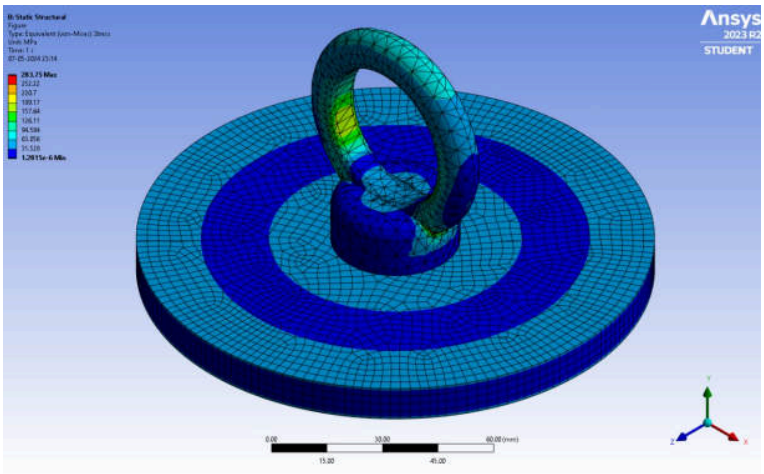


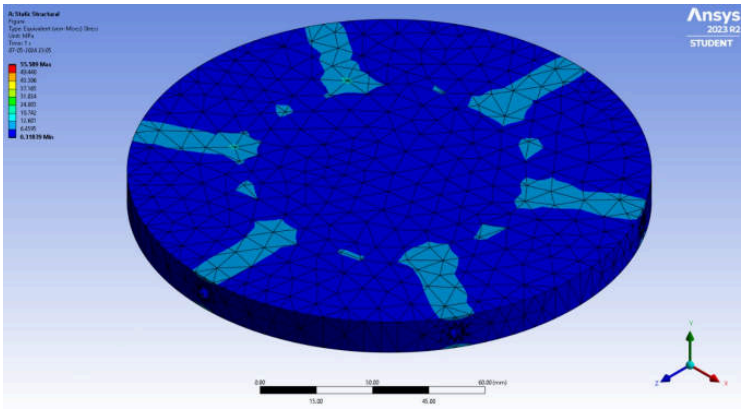


3. Bulkheads

DESIGN:
The design of the bulkhead is straightforward and simple. Bulkheads are important structural members that help transfer loads to the airframe. The motor and recovery deployment are the major load causing parts. The bulkhead that is connected to the Parachute and shock cord is made of Fiberglass and is 0.5 inch thick. It is firmly screwed onto the upper body tube.
The motor bulkhead or thrust plate employed a similar philosophy of tested and simple design. Aluminum alloy (6061) was chosen for the thrust plate because of its superior strength and ability to withstand the loads that are generated by the motor.

MANUFACTURING:
The fiberglass bulkhead is made using wet lay-up. The detailed process is described later in this report.
The Aluminum thrust plate was CNC cut out of a 0.5 inch thick plate.

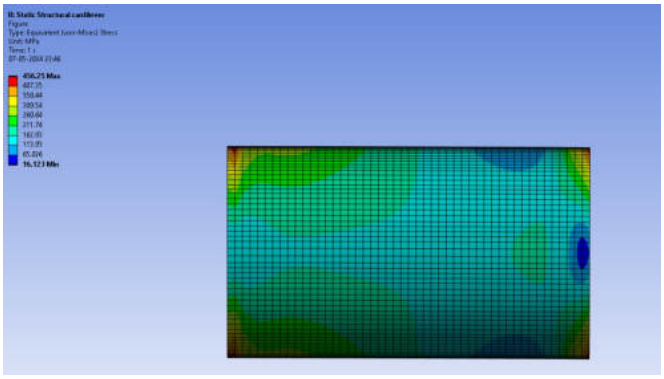




4. Coupler

The length of the coupler was chosen by convention as to be double the radius of the fairings (Body tubes), to avoid and minimize any buckling. The coupler is screwed onto one of the two airframes using M4 screws. Since recovery will be separating the two body tubes, shear pins are added

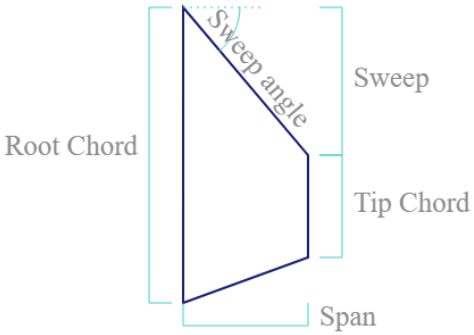
between the coupler and the lower body tube. Fiberglass is again used to manufacture the coupler through the wet layup process.



5. Fins

DESIGN:
The design of the fins is optimized to provide stability throughout the flight. The target is to reach maximum apogee while producing the least amount of drag
OpenRocket was used to compare three different plan form shapes:
1. Swept back profile
2. Symmetrical trapezoidal
3. Clipped delta

To compare these profiles fairly one parameter is kept constant, that being area. Hence, keeping the planform area constant, other parameters are arranged to get optimized fins. The swept back profile produces best results with maximum apogee and least drag at maxQ, the stability margins of at least 2 throughout the ascent is maintained while designing the fins.



A rounded cross section is chosen for the fins as it reduces drag and increases apogee. While directly manufacturing rounded cross section fins was not possible through wet layup method, initially a rectangular cross section was taken, and the corners were sanded smoothly into a round cross section.
Calculations were performed to calculate the flutter velocity using equation from Apogee components newsletter:

1. Flutter velocity:

$$V_f = a\sqrt{\frac{G}{\frac{1.337AR^3P(\lambda+1)}{2(AR+2)(\frac{t}{c})^3}}}$$

2. Surface area:

$$S = \frac{1}{2}(Cr + Ct)b$$

3. Aspect ratio and lambda:

$$AR = \frac{b^2}{S}$$

$$\lambda = \frac{Ct}{Cr}$$

4. Temperature and pressure variations:

$$T(°F) = 59 - .00356h$$

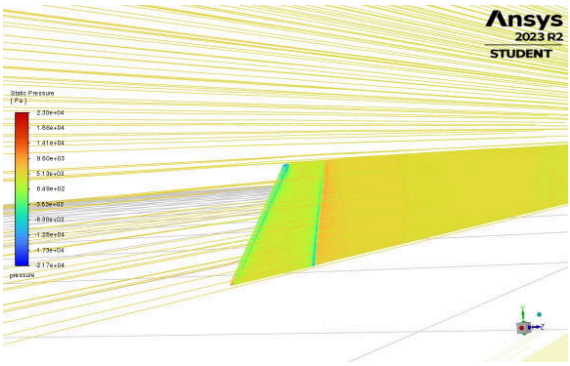
$$P(lbs/ft^2) = 2116 \times \left(\frac{T + 459.7}{518.6}\right)^{5.256}$$

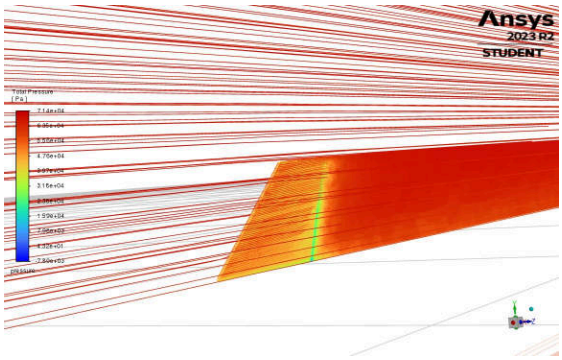
FIN DIMENSIONS:

Parameter	Value (in.)
Root Chord	7.874
Tip chord	3.937
Sweep Length	5.906
Height	4.528
Thickness	0.118

MANUFACTURING:
The main purpose of fins is to stabilize the ascent of the rocket during and prevent any spin of the rocket. In the whole duration of flight, aggressive aerodynamic forces are acted on the fins. Agneya is equipped with 4 fins which are equally spaced around the airframe and mounted on the lower body tube.

Fin flutter is a major problem when it comes to high velocities. In this case the maximum velocity attained by the rocket is 920 ft/sec. Fin failure can cause problems during the flight. To prevent that it is necessary that the designed fins pass the flutter survivability. The material chosen for the fins is glass fiber and wet layup is the manufacturing technique used.
The GFRP fins were made in the form of plates. The glass fiber plates are made with the required thickness of 0.118 inches. These glass fiber plates are then cut into the fin profiles with water jet cutting. The fins will be attached to a structure called fin can which also acts as motor retainer.





D. Payload

A) OVERVIEW:

Our payload design integrates a specialized quick-reaction stabilization gimbal system tailored specifically for the demanding conditions encountered during research and development on sounding rockets. Its primary objective revolves around ensuring the preservation of sensitive payloads within the confined space of a rocket's payload section. Essentially, the payload functions as a meticulously engineered stabilized platform, aimed at counteracting the significant forces experienced during the flight of sounding rockets and providing active stabilization.

TECHNICAL INFORMATION:

1. Mechanical Design: The gimbal bay will feature a 3U CubeSat Structured Payload fitted with gimbal BLDC motors, positioned to facilitate precise control over the platform's orientation.

2. Control System: Operating on a PID closed control loop, the stabilizer system will continuously monitor and adjust gimbal angles to maintain the platform's stability throughout the rocket's trajectory. The integration of magnetic encoders (AS5600) will enrich the feedback loop, enhancing the system's responsiveness and adaptability to dynamic flight conditions.

3. Data Collection: The payload will be equipped with sensors to collect data on the performance of the stabilizer system throughout the flight. This data will include IMU data, gimbal joint angles, and environmental conditions, providing valuable insights for future optimization and refinement.

This payload design represents a significant advancement in our stabilization technology, expanding upon the foundations laid by our previous Passive Vibrational Attenuation Device (PVAD) payload. By offering a versatile and actively controlled platform capable of accommodating shock-sensitive payloads, it ensures the

integrity of the cargo, mitigating risks associated with the high impulse flight of sounding rockets. Additionally, the payload integrates stable power systems and a Data Acquisition and Control System (DAC), enabling seamless interaction and data collection, thereby enhancing overall mission effectiveness.

In summary, our payload design provides a technical solution to the challenges posed by research and development on sounding rockets. With its focus on precise and rapid orientation control and stability, it ensures the safeguarding of shock-sensitive payloads, thereby contributing to the success of scientific endeavors in this field. Furthermore, adherence to design guidelines and regulations ensures that the payload maintains a mass of approximately 8.6 lbs, although this figure may vary due to mass estimations derived from CAD software. Importantly, the payload does not contain hazardous or radioactive materials, nor does it include any form of pyrotechnics.

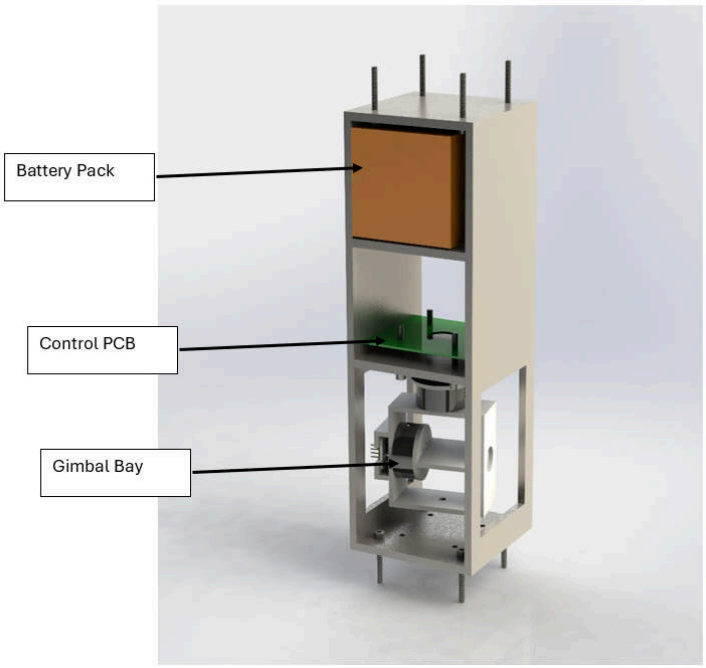


Fig: Payload prototype render

B) COMPONENTS:

Component	Quantity	Function
ESP32	1	Microcontroller
L298N	2	Motor Driver
Gimbal BLDC 140kV 2805	2	Gimbal Motors

AS5600	2	Magnetic Encoders for Closed Control Loop
AMS1117	1	Voltage Regulator
LM2596	1	Buck Convertor to Step-down
LSM6DSO	2	IMU for stabilized and non-stabilized bays
MPU6050	2	Redundant IMU
SD-Card Module	1	SD Card Interface Module for Primary Storage
W25Qn	1	Flash Chip Module for Redundant Storage
WS2812	1	RGB LED State Indicator
Connectors	TBD	Interconnection between all units
Terminal Blocks	TBD	Connection of Battery and Gimbal to PGC
4S3P Li-ion Battery Pack	1	Battery Pack

The Payload Gimbal Controller (PGC) is designed to effectively control the orientation of a rocket payload. Utilizing the ESP32-WROOM chip, selected for its affordability, appropriate input/output capabilities, and adaptability, the PGC integrates several key functionalities.

Firstly, the power distribution section facilitates the connection of the battery and other essential components. It incorporates connectors and mechanisms for voltage regulation, ensuring stable power delivery to the microelectronics within the system. The Voltage distribution has been demonstrated in the image below –

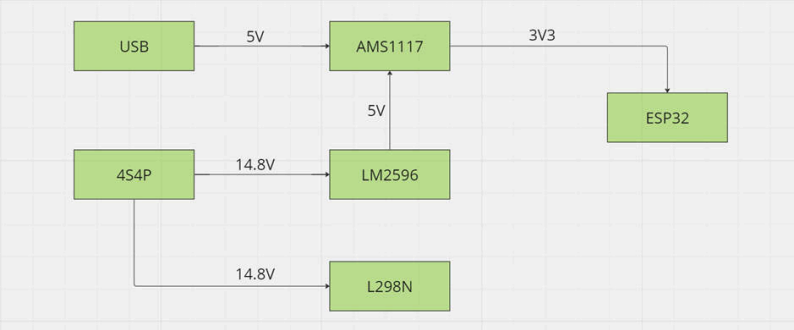


Fig: Power distribution flowchart

Secondly, the gimbal controller section features the driver, IMU, and connectors for motor and encoder wiring. This enables precise control over the gimbal's movements, allowing it to adjust in response to the rocket's orientation.

Additionally, the PGC includes components for monitoring and logging data. It tracks the state of the PGC Firmware and compares data from the LSM6DSO stabilized IMU (sIMU) and the onboard IMU (oIMU). This comparative analysis assesses the effectiveness of the gimbal in stabilizing the rocket's orientation.

Data logging primarily occurs on an SD card, providing a robust storage solution. However, to mitigate the risk of data loss due to shocks or vibrations, redundant storage on a Flash Chip is implemented. This ensures data integrity even in the event of SD card disconnection.

C) METHODOLOGY

To enable closed-feedback control loop operation, we utilize AS5600 magnetic encoders strategically positioned behind the motor's shaft. These angles are then input into our Inverse-Kinematics (IK) equations to drive the BLDC motors. Control of the BLDC motors is facilitated by a modified 2-channel L298 motor driver, configured to output 3-PWM output channels for motor operation. The control flow is implemented through a closed control loop with the callback data from the magnetic encoder and is as follows –

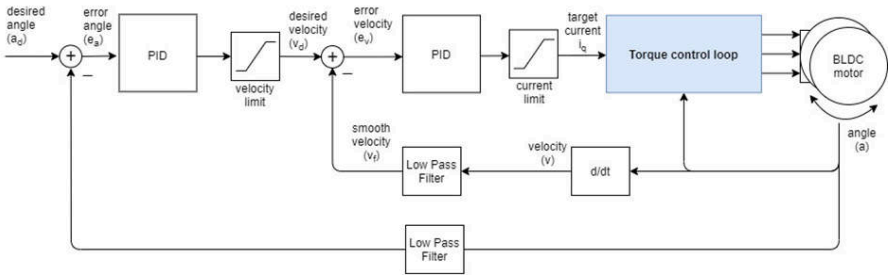


Fig: Closed control loop- position hold control flow

Evaluation of payload effectiveness will be conducted through a two-parameter comparison approach: first, by deploying an IMU on the stabilized platform with appropriate filters to capture data, and second, by interpolating values from the magnetic encoder to calculate the absolute orientation of the end-effector.

D) DEVELOPMENT AND TESTING

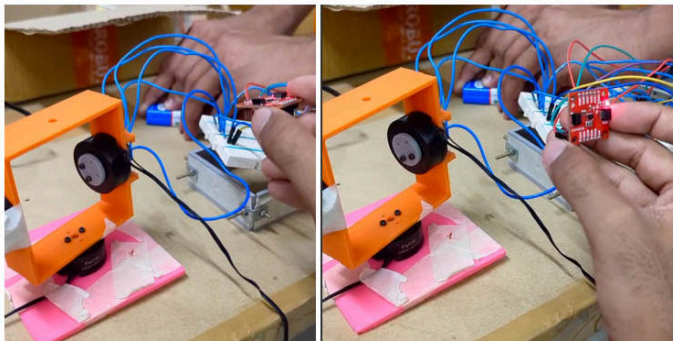


Fig 4: Implementation of a p-controller with IMU orientation data

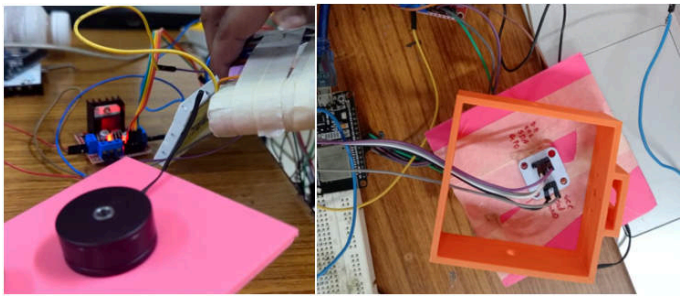


Fig 5: a) L298N Load test with 6S Li-ion Battery Pack
b) Positioning and Calibration of AS5600 Magnetic Encoder

E. Avionics Subsystems

AVIONICS BAY:

The team's rocket Agneya, will feature an avionics bay with a 3D-printed structure that securely holds:

1. Two COTS Altimeters/Flight Computers
    a. MissileWorks RRC3 (Primary)
    b. Featherweight Raven 4 (Secondary)
2. One COTS GPS Tracker: Featherweight GPS Tracker
3. Two SRAD Altimeters/Flight Computers: SPARC 4
4. One SRAD GPS Tracker: Bessie

Battery bay:
The empty space on the inside of the "Triangular Prism" shaped avionics bay will be utilized as the battery bay. The battery bay will be an independent module which will be capable of sliding in and out of the avionics bay and will house all the necessary batteries for the entire avionics systems. The batteries will be quick-swappable for easy and rapid exchange.
The batteries involved are:

1. 3x standard 9V batteries (from Duracell/Energizer or other reputed brands) for both COTS Altimeters/Flight Computers and an additional battery for redundancy).
2. 3x 2S (7.4 V) 18650 Samsung Li-ion batteries for the three SRAD components.
3. 600mAh Li-Po battery for COTS Featherweight GPS Tracker (*special case: recommended by the manufacturer)

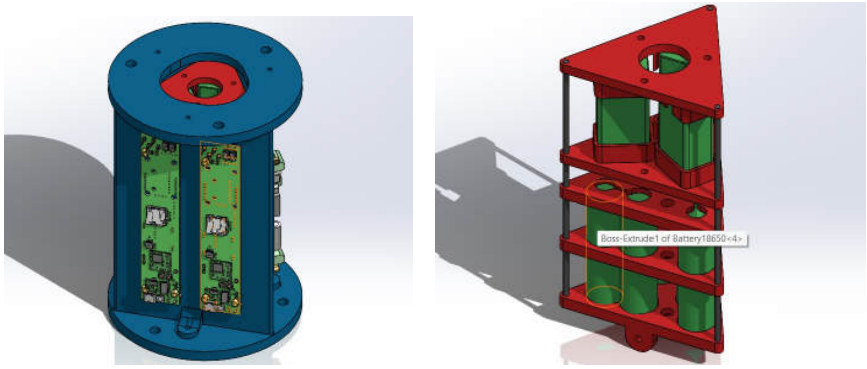


Fig: a) Avionics bay b) Battery bay (to go inside the avionics bay)

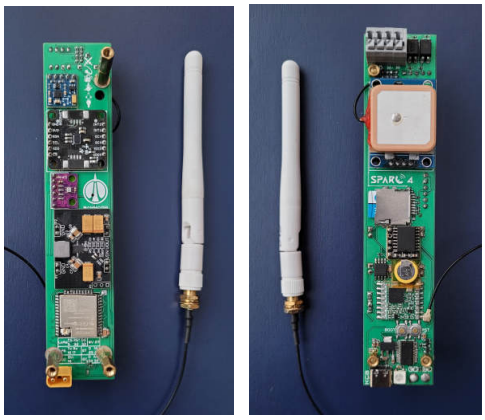
The avionics bay is in the shape of a “Triangular Prism”, on one of the faces, the two primary and secondary COTS Altimeters/Flight Computers will be placed involved in recovery. These flight computers will play a crucial role in accurately measuring the rocket's altitude during its flight.

The second faceplate within the avionics bay will house two SRAD Altimeters/Flight Computers, SPARC 4 for data collection and telemetry. SPARC will serve as a payload specifically intended for testing, research, and development purposes for our potential SRAD category flights in the future. It should be noted that the SRAD Altimeters will not be involved in the recovery procedure of the rocket. Instead, its primary function will be to gather data and measure the accuracy of the team's intermediate flight software implementation.

The data collected from this SRAD Altimeter will provide valuable insights for the ongoing development of the team's dedicated SRAD flight computer, which will be utilized in future launches. This iterative approach will enable the team to refine and enhance the performance of the SRAD flight computer, ensuring its effectiveness and reliability in upcoming missions.

The third and final faceplate will support the two COTS and SRAD GPS trackers, Featherweight GPS Tracker and Bessie, respectively. These GPS trackers will aid search and recovery of the rocket upon the conclusion of the flight.

SPARC 4, SRAD Flight Computer:
SPARC 4 is a fourth generation Flight computer developed by Team Sammard for the SRAD Altimeter/Flight Computer component. It consists of BMP280, a highly reliable and accurate pressure sensor being used as an altimeter here with ±1 meter accuracy (3.28 feet), a very reliable GPS provided by U-blox, the NEO-6M GPS module, which provides ±2.5 meter (8.2 feet) positional accuracy, a 6-axis Inertial Measurement Unit LSM6DS0 Accelerometer and Gyroscope and HMC5883L Magnetometer for obtaining the physical parameters and calculating the attitude of the rocket. The telemetry acquired from the sensors and calculations is transmitted over 915MHz LoRa RFM95W. Using LoRa Communication gives us immense range, sufficiently enough for the 10K COTS category.
All the data obtained and calculated by SPARC is logged onto an onboard microSD card and an SPI Flash chip simultaneously with proper time stamps acquired from an independent on-board RTC so that the data can be recovered upon the accomplishment of the flight as well. In addition to these capabilities, SPARC also includes trigger detection and continuity measurement for the e-matches connected to the primary COTS flight computer, MissileWorks RRC3.
A second SPARC flight computer will also be placed alongside as a back-up data logger for the flight



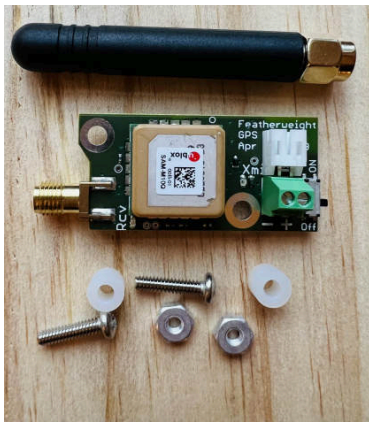




Fig: a) SPARC 4 b) Featherweight COTS c) RRC3

COTS GPS, Featherweight GPS Tracker:
The Featherweight GPS Tracker will relay the position of the rocket to the ground station in real-time with the precise location of the rocket in terms of latitude, longitude, and altitude AGL. A ground station endpoint which is handled by the user is responsible for relaying the data to a mobile phone. The relevant GPS data (latitude, longitude, and altitude) will be relayed from the COTS GPS tracker inside the rocket to a device on the ground station over LoRa protocol in the 915MHz frequency band (adjustable) and is displayed on a mobile application developed by Featherweight via Bluetooth. The app also shows an acknowledgment of a successful establishment of a connection between the application and the ground station. Featherweight GPS tracker was also tested for its satellite fixes available. As these components would be placed in high-powered rockets, efficiency should be maintained throughout the duration of the flight. The Featherweight GPS tracker was tested thoroughly, and the data provided by the GPS tracker was cross verified with other commercial GPS sources. The adequate number of satellite fixes was also one of the various other positive results gained from the testing.

SRAD GPS, Bessie:
The SRAD GPS tracker, Bessie, is a first-generation GPS Tracker developed by Team Sammard for the SRAD GPS component. It incorporates a U-blox NEO-6M GPS module for a great ±2.5 meter (8.2 feet) positional accuracy. The GPS telemetry consisting of latitude, longitude, altitude, speed, heading, and satellite fix is transmitted to a ground station receiver module “Shrota” over 915MHz LoRa RFM95W connected to the teammate’s mobile phone running a custom GPS tracker application developed by Team Sammard itself.

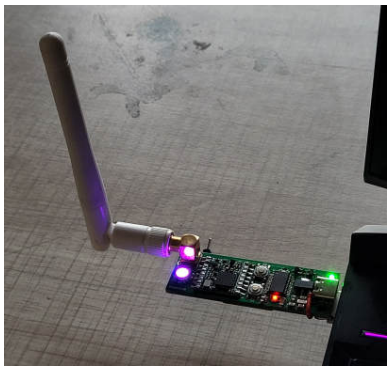
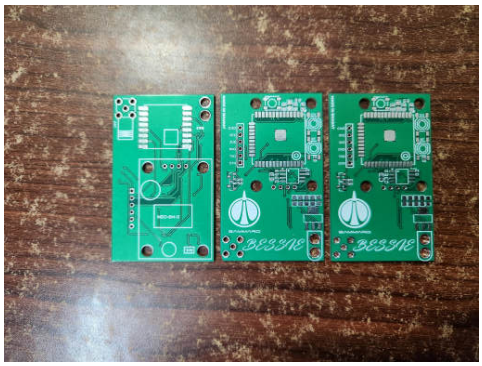


Fig: a) Shrota



b) Bessie

*All batteries involved are in accordance with the latest battery guidelines provided by ESRA.

F. Recovery

PRIMARY DEPLOYMENT SYSTEM
The primary system contains the MissileWorks RRC3 altimeter and the secondary system contains the FeatherWeight Raven 4 altimeter. One 9v battery and two 3.7v 2P battery provide power to these altimeters. The primary system is programmed to deploy the drogue parachute at the apogee, and the main parachute is programmed to deploy at an altitude of 1,490 ft. The secondary system is programmed to deploy the drogue after a set delay of a few seconds after reaching apogee and then the main parachute after a set delay of a few seconds after reaching 1,500 ft. This is achieved by using Black Powder mechanism. Both the primary and secondary altimeters control two ematches each (drogue and main) which are tunneled to the flange of the corresponding charges. In case of failure of either one of them, the redundant mechanism triggers it in

accordance with the delay logic mentioned above.

REDUNDANT DEPLOYMENT SYSTEM
A redundant mechanism is incorporated into the design to ensure parachute deployment in case of primary mechanism failure. Seven grams of FFFFg black powder are used for redundant charges. The secondary system contains the Featherweight Raven altimeter and is programmed to deploy the drogue after a set delay of a few seconds after reaching apogee.
The entire assembly was machined in-house, allowing for customization to increase reliability. The final machined parts were given a liner of Plaster of Paris. The structure was machined from Aluminum.

TENDER DESCENDER
The tender descender is attached to the tether between the drogue and the eyebolt and holds the main parachute. It also uses a black powder separation with its redundant mechanism and deploys the main parachute at an altitude of 1500 feet. According to weight calculations and verifications, the approximate force required for tender descender separation was 180 N. According to black powder calculations, the mass of black powder required to achieve the force was approximately 0.9 grams, with a significant safety factor.

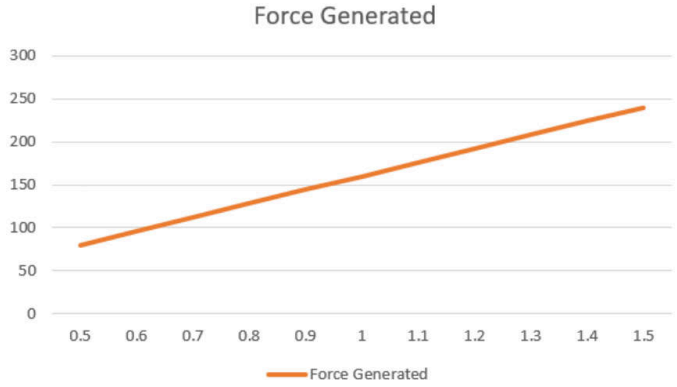
Due to restrictions in black powder purchasing in India, FFFFg black powder could not be used for testing. The team developed their own in-house black powder to overcome this difficulty. This was done using charcoal, sulphur, and potassium nitrate in a fixed ratio. Recovery tests have been conducted successfully using the same black powder.





Fig: a) Recovery mechanism b) Tender Descender mechanism

For FFFFg calculations, forces were verified experimentally and black powder equations were used to calculate the mass required. A fuel characterization graph was obtained to compare our in-house manufactured with the standard FFFFg black powder.



G. Flight Software

Design:
The flight software for Agneya rocket is developed in embedded C using the Arduino IDE, adhering to the specifications of the SPARC flight computer for which the program is designed. An ESP32 microcontroller serves as the SPARC flight computer, and the program leverages various sensor and storage libraries, including the EEPROM library, LSM6050 library, HMC5883L library, NEO-6 M library, RTC (Real Time Clock) library, SimpleKalmanFilter library, math library, Wire library and the SD card library provided by the IDE. The BMP280 sensor integration was achieved solely through code configurations, without reliance on external libraries.
Additionally, the software employs the SPI library, and SPI flash storage for faster data access compared to SD card operations. The sensor libraries are utilized to receive sensor data inputs, while the EEPROM library is employed to persistently store essential data as a backup, ensuring data retrieval and continued processing in the event of power loss. The SD library is used to save data to a micro SD card for later extraction.
In order to mitigate the significant time overhead associated with file manipulation processes like opening, appending data, and closing files on an SD card, the decision was made to utilize SPI flash storage due to its notably faster performance. However, as a precautionary measure, backup data continues to be stored on the SD card. Upon powering on the SPARC flight computer, packets of data are generated from the sensors and stored in CSV files on both the micro SD card and SPI flash at a rate of 1Hz.
The flight states are triggered via conditional statements, and any sensor reading errors due to noise are accounted for by the SimpleKalmanFilter library.

The default state upon powering on is state 0, the pre-ignition state, used to calibrate the sensors and prepare the flight computer before launch. Once the processor detects an increase in altitude and change in acceleration, the state transitions to state 1, the ascent state.

As the rocket reaches apogee, the velocity decreases until the instantaneous altitude value ceases to be greater than the previous packet's altitude value. If this condition is met, the state changes to state 2, the descent state.

During this descent state, the flight computer executes a continuity check at the apogee to determine whether the drogue parachute has been deployed.

During state 2, packets are recorded as normal, and the altitude is continuously monitored. At 1500 ft another continuity check is performed to determine if the main parachute has been released. If the altitude is confirmed to not be reducing further after a point, the state changes to state 3, the landing state.

State 3 is the final state, where, upon detecting appropriate altitude and acceleration conditions, the buzzer is activated to signal the rocket's landing, and a landing alert is written into the file.

Throughout the duration of program execution, data packets are generated, stored, and transmitted to the ground station FLAIR at a rate of 1Hz over a LoRa module operating at 916MHz frequency. This ensures both the linear recording of data and the real-time monitoring of conditions for proper program execution. Additionally, the flight computer will transmit GPS data to Shrota, which will be accessible at the ground control station and through a mobile app, enabling data-visualization and post-landing rocket tracking.

Utilization:
The flight software serves as a vital part of the Agneya rocket system, handling real-time sensor data to carry out crucial tasks during the flight. Its main roles include gathering sensor readings, processing them into usable packets, and managing communication with ground stations and Shrota through telemetry. While it currently manages sensor calibration and flight state changes, future updates will enable parachute

deployments. By detecting launch, peak altitude, and landing, the software ensures the mission progresses smoothly and safely, highlighting how embedded C programming and advanced hardware in the SPARC flight computer work together effectively.

H. FLAIR – SRAD GROUND CONTROL SOFTWARE

Overview:

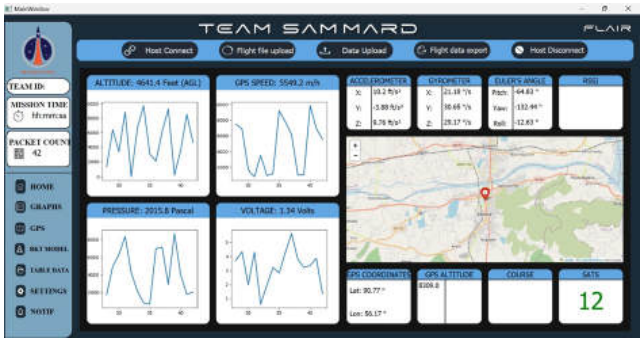
FLAIR, our Ground Control System meticulously crafted for the Agneya sounding rocket mission, is intricately tailored to our specific requirements. Developed utilizing Python and PyQT libraries, FLAIR, known as the Flight Analysis Interface, seamlessly integrates into our operations by receiving real-time data from the SPARC flight computer via Shrota. Beyond its primary function of data reception, FLAIR assumes a pivotal role in both visualizing and securely storing data. Its capabilities extend to real-time data plotting and storage, both onboard the flight and on the ground, ensuring the preservation of crucial information.

FLAIR's intuitive interface presents data in visually compelling line graphs, empowering our team to comprehensively understand data behavior and accurately predict trends. Moreover, it not only facilitates real-time data plotting but also guarantees the safe storage of data on the ground system. This aspect is paramount for maintaining a comprehensive record of mission data, facilitating post-mission analysis for invaluable insights and future launch enhancements.

In essence, FLAIR's archived data serves as a repository of knowledge, enabling our team to refine rocket designs, optimize performance, and achieve mission objectives with heightened efficiency and precision.

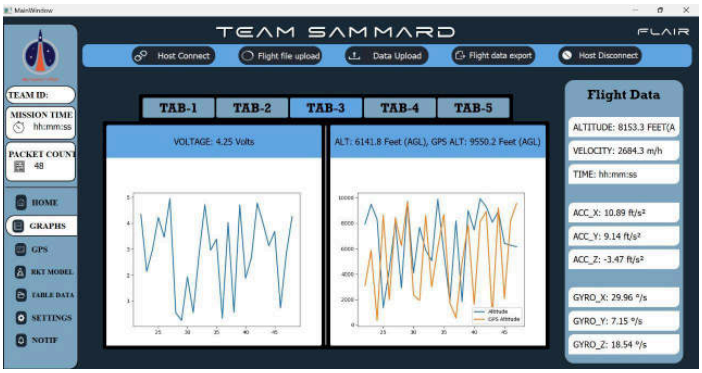
Various functionalities in FLAIR:

1. **Home Page:**
   The homepage is essentially our dashboard which features four essential graphs tracking altitude versus time, GPS speed versus time, pressure versus time, and voltage versus time, offering a clear understanding of your rocket's orientation throughout its flight. Additionally, we also display crucial flight parameters such as accelerometer and gyrometer data, Euler angles, RSSI, GPS coordinates, altitude, course, and satellite information. These graphs and table data facilitate easy visualization and analysis of your rocket's performance during every phase of its mission. Moreover, our GPS location display assists in locating and recovering the rocket after its flight.

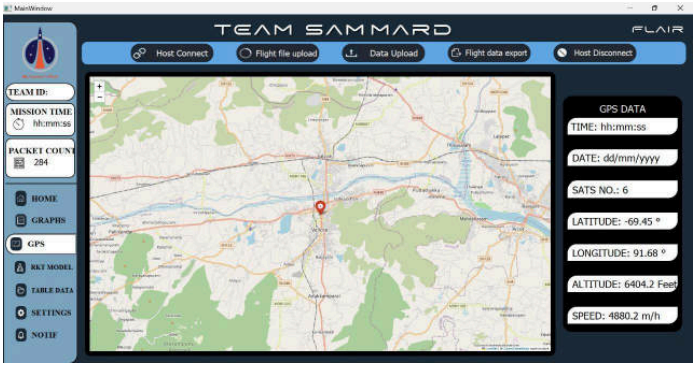


2. **Graphs Page:**
   The Graphs page serves as a pivotal hub, featuring five tabs, each meticulously curated to provide real-time insights. Within each tab, users can explore two graphs, including essential metrics such as Pressure vs. Time, Temperature vs. Time, Velocity vs. Time, GPS Altitude vs. Time, Accelerometer vs.

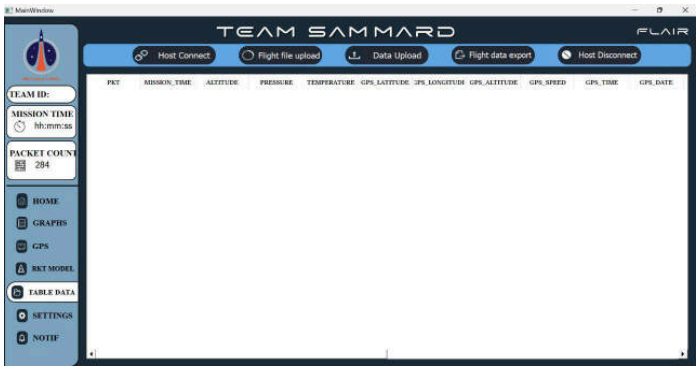
Time, Gyrometer vs. Time, and GPS Speed and Velocity vs. Time. These graphs, prominently displayed for efficient visualization, offer a detailed overview of flight dynamics throughout the journey. Additionally, the Flight Data panel offers a succinct display of crucial flight parameters, including altitude, speed, and time, alongside acceleration along the X, Y, and Z axes, as well as gyrometer values along the X, Y and Z axes.



3. **GPS Page:**
The GPS page is for efficient rocket recovery. With an enlarged GPS display, locating your rocket becomes a breeze, ensuring swift and precise retrieval. The GPS data panel provides crucial parameters including time, date, number of satellites, latitude, longitude, altitude, and speed, offering comprehensive insights for seamless monitoring and management.

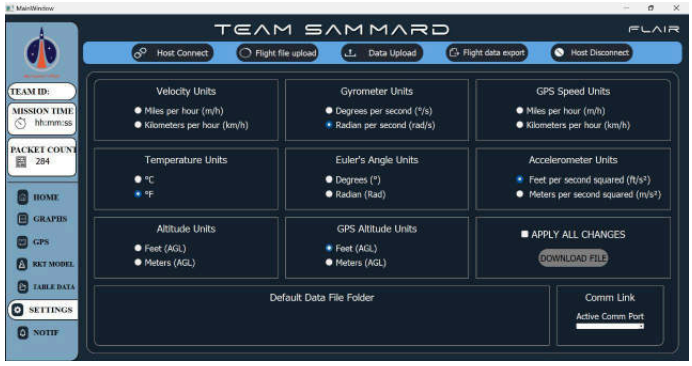


4. **Table Data Page:**
The Table Data Page is a comprehensive interface for accessing flight parameters from the flight software system. It has a tabular display showcasing all the crucial data received, both during and after flight. Whether monitoring in real-time or conducting post-flight analysis, every data point is readily available for scrolling and in-depth analysis.



5. Setting Page:

The Settings page is the hub for managing unit conversions and comport information. It facilitates the conversion of various flight parameters in the saved CSV file, generating a new CSV file with updated units. Once selected the required units and clicked on the APPLY ALL CHANGES button, conversions are swiftly executed, and the updated CSV file is promptly saved. Additionally, the Comm Link section provides clear visibility of the active comm port used for established connections.



Top Panel Buttons:

a. HOST CONNECT BUTTON:

The "Host Connect" button facilitates the establishment of an open connection between the communication port in the ground control system and the rocket's flight software.

b. FLIGHT FILE UPLOAD BUTTON:

The "Flight File Upload" button initiates the process of reading data from a comma separated value file line by line and subsequently plotting it on graphs, synchronized with time.

c. FLIGHT DATA EXPORT BUTTON:

The "Flight Data Export" button allows users to navigate the system and save the generated comma separated value file to a designated location after the end of flight.

d. DATA UPLOAD BUTTON:

The "Data Upload" button displays plots created from comma separated value file values, though without real-time synchronization. This button is used to display flight trajectory analysis by sidelining graph plotting.

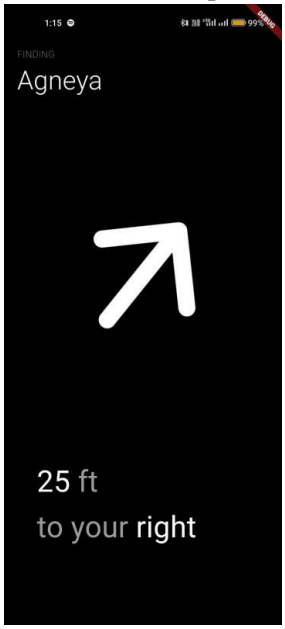
e. HOST DISCONNECT BUTTON:
The "Host Disconnect" button terminates the communication port connection, thereby halting the reception of data from the flight software and making it available for other uses.

In conclusion, FLAIR is an advanced and powerful Ground Control System that plays a role in the launch of our sounding rocket, AGNEYA. It provides us with real-time telemetry data and allows us to store this data for future analysis and improvement. With its user-friendly interface and visually appealing line graphs, FLAIR helps our team to understand the behavior of the data and improve the performance of our further flights on the basis of the data.

Rocket Tracking Application

As part of the Agneya rocket mission, our team has developed a cross-platform mobile application using Google's Flutter framework. This app is designed to assist the recovery team in locating and tracking the rocket's position after it has landed. It seamlessly operates on both iOS and Android devices, providing a consistent user experience across multiple platforms. The app receives real-time GPS data from Shrota, which is fed by the rocket's flight computer during the mission. This data along with the current gps coordinates and the inbuilt compass in the phone help us determine which direction the rocket is in and how far it is.

To further aid the recovery process, the app calculates and displays the distance between the user's current location and the rocket's landing site, and also points towards the rocket. This information enables the team to plan their approach efficiently and allocate resources accordingly. The user interface of the app is designed to be intuitive and user-friendly, ensuring that the recovery team can quickly access the necessary information without any unnecessary complexities. Real-time updates and advanced tracking capabilities ensure that the team stays informed throughout the retrieval process.



I. Manufacturing

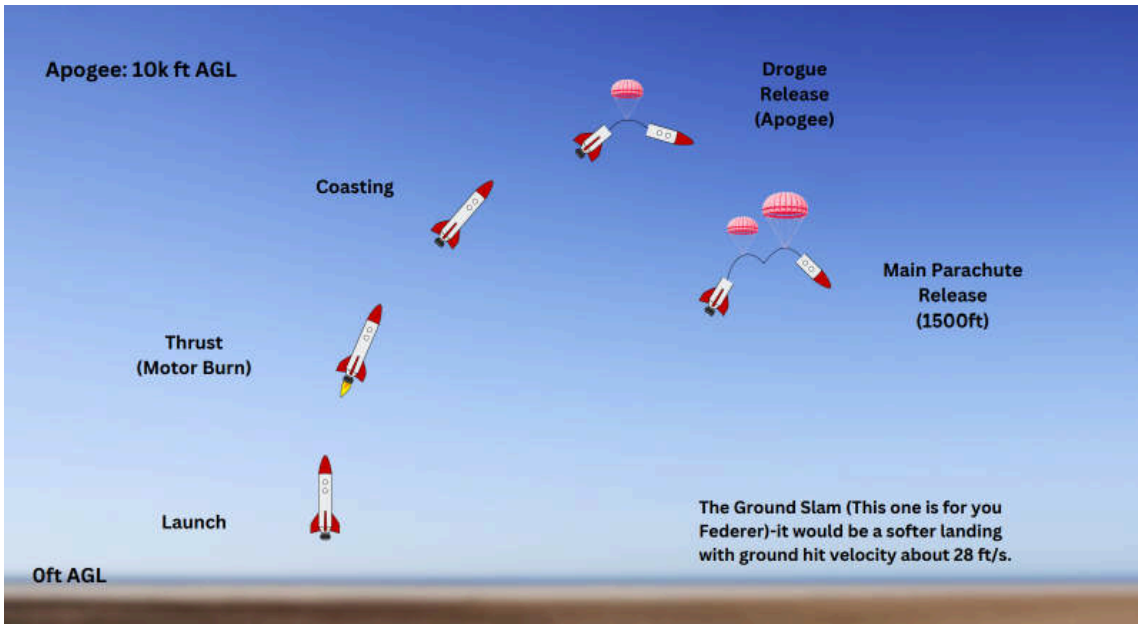
Vacuum bagging technique was used to fabricate fiberglass plates. After calculating the number of fiberglass sheets required to achieve, wet layup was done using Epoxy. The fabric to resin ratio for our fiberglass is 1:2 and for the resin the epoxy to hardener ratio is 10:1. To get proper mix of the epoxy we need to stir it for atleast 5 mins and when the mixture is moderately warm on touching is the time at which we start the layering of the fiberglass. Curing time for the resin is 15-20 mins so we need to do layering of the fiberglass looking at curing time. The fiberglass surface treatment, fabric to resin ratio and the impregnating technique are some of the very crucial fabricating aspects that must be carefully considered and observed when producing high performance composites.

For 3D printing various components, we used the Creality ender 3 v2 neo printer. Depending on the requirement in the strength of the component, ABS or PLA plastic filament is used. Components such as the cup for the recovery of the rocket, tender descenders, avionics bay were printed.



Fig: Vacuum bagging and manufactured fiberglass plates

III. MISSION CONCEPT OF OPERATIONS



IV. CONCLUSION AND LESSONS LEARNT

Reflecting on the challenges and triumphs of the past year, the Team embarks on a new chapter with the project "AGNEYA" for the upcoming IREC Spaceport America Cup. After the successful launch and recovery of Vajra last year, the team is poised to leverage the lessons learned and experiences gained to propel AGNEYA to new heights.

Design Evolution:
AGNEYA marks a significant evolution from Vajra, presenting fresh design challenges and opportunities. With a focus on scalability and innovation, the team is tasked with optimizing the rocket's performance and functionality within the constraints of the competition requirements.

Learning from Vajra's success and shortcomings, the team approaches AGNEYA with a refined understanding of rocket integration, component sizing, and payload capacity. The journey of AGNEYA represents a continuous quest for excellence and exploration in the realm of experimental rocketry.

Manufacturing Advancements:
Building upon the lessons gleaned from Vajra's manufacturing process, the team embraces new techniques and technologies to enhance efficiency and precision. Leveraging both in-house capabilities and external resources, AGNEYA's components are crafted with meticulous attention to detail and quality.

The iterative nature of manufacturing is embraced, recognizing that failures are stepping stones to success. Through perseverance and collaboration, the team overcomes obstacles and refines their craftsmanship, setting a higher standard for AGNEYA's construction.

Team Dynamics and Knowledge Transfer:
Assembling a diverse team of experienced seniors and eager sophomores, the AGNEYA project fosters a culture of continuous learning and mentorship. Nights in the lab are filled with spirited discussions, collaborative problem-solving, and hands-on experimentation.

Sophomores are immersed in the fundamentals of rocketry, guided by the wisdom and expertise of their senior counterparts. Knowledge transfer is facilitated through interactive sessions, tests, and presentations, ensuring that each team member contributes to AGNEYA's success.

Conclusion and Anticipation:
As the countdown to the IREC Spaceport America Cup begins, the team reflects on the collective journey of perseverance, growth, and discovery. Memories of late nights in the lab are interwoven with the anticipation of AGNEYA's debut on the competition stage.

AGNEYA represents more than a rocket; it embodies the culmination of months of dedication, innovation, and teamwork. With excitement and determination, the team looks forward to not only witnessing AGNEYA's performance but also to embarking on a new chapter of exploration and achievement in rocketry.

SYSTEM WEIGHTS, MEASURES, AND PERFORMANCE DATA APPENDIX

A) BASIC ROCKET INFORMATION:

Parameter	Value	Units
Vehicle length	85.236	in
Airframe diameter	5.906	in
Vehicle weight	572	oz
Propellant weight	133	oz
Empty casing weight	103	oz
Payload weight	140	oz
Liftoff Weight	947	oz
Center of pressure	59.856	in
Center of gravity	48.06	in
Number of fins	4	-
Fin height	4.528	in
Root chord	7.874	in
Tip chord	3.937	in
Fin thickness	0.118	in

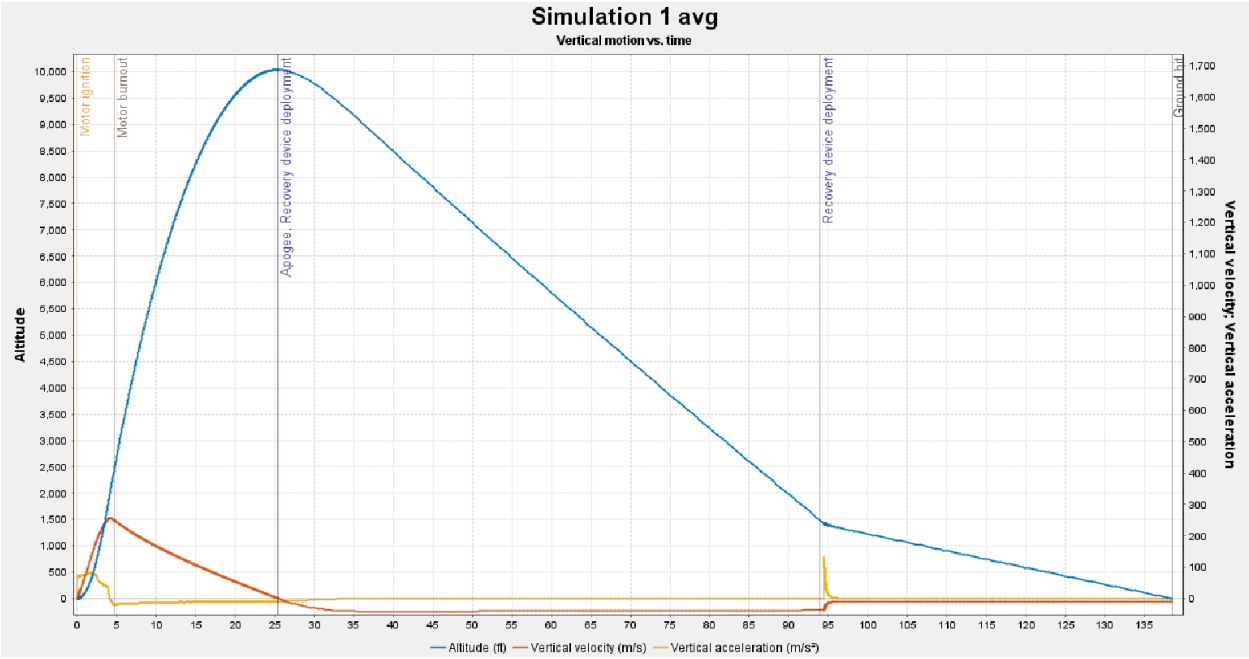
B) PROPULSION INFORMATION

Parameter	Value
Motor name	Aerotech M1845NT-PS
Burn time	4.4 s
Average thrust	1822 N
Total impulse	8093 Ns

C) PREDICTED FLIGHT DATA

Parameter	Value	Units
Launch rail length	17	ft
Liftoff thrust-weight ratio	9.21	-
Rail departure velocity	90	ft/s
Minimum static margin	2	cal
Maximum acceleration	8.64	G
Maximum velocity	865	ft/s
Fin flutter velocity	920	ft/s
Target apogee	10000	ft
Predicted apogee	10154	ft

D) FLIGHT PROFILE GRAPH



E) RECOVERY INFORMATION

Parameter	Value
Primary Altimeter	Missileworks RRC3
Redundant Altimeter	Featherweight Raven
Drogue primary deployment charge	5.2 g
Drogue redundant deployment charge	5.8 g
Main primary deployment charge	0.9 g
Main redundant deployment charge	1.1 g
Drogue deployment altitude	10154 ft (Apogee)
Drogue descent rate	123 ft/s
Main deployment altitude	1490 ft
Main descent rate	31 ft/s

PROJECT TEST REPORTS APPENDIX

1. SRAD Propulsion System Testing
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2. Combustion Chamber Testing
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3. Hybrid and Liquid Propulsion Tanking Testing
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4. Static Hot-Fire Testing
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5. SRAD Pressure Vessel Testing
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6. Proof Pressure Testing
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7. Optional Burst Pressure Testing
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8. Recovery System Testing
The team has tested using body tubes and couplers.

8.1 Ground test demonstration

Drogue:
The system consists of a double chamber that contains a primary and redundant charge, and a compression fit to hold the cup structure inside the primary chamber in. An e-match is fed through a small hole on the bottom side of the both redundant chamber and the cup structure of the primary chamber and then sealed off to prevent pressure leaks.

Main:
The tender descender is attached to the tether between the drogue and the eyebolt and holds the main parachute. It also uses a black powder separation with its redundant mechanism and deploys the main parachute at an altitude of 1490 feet. According to black powder calculations, the mass of black powder required to achieve the force was approximately 0.9 grams, with a significant safety factor.

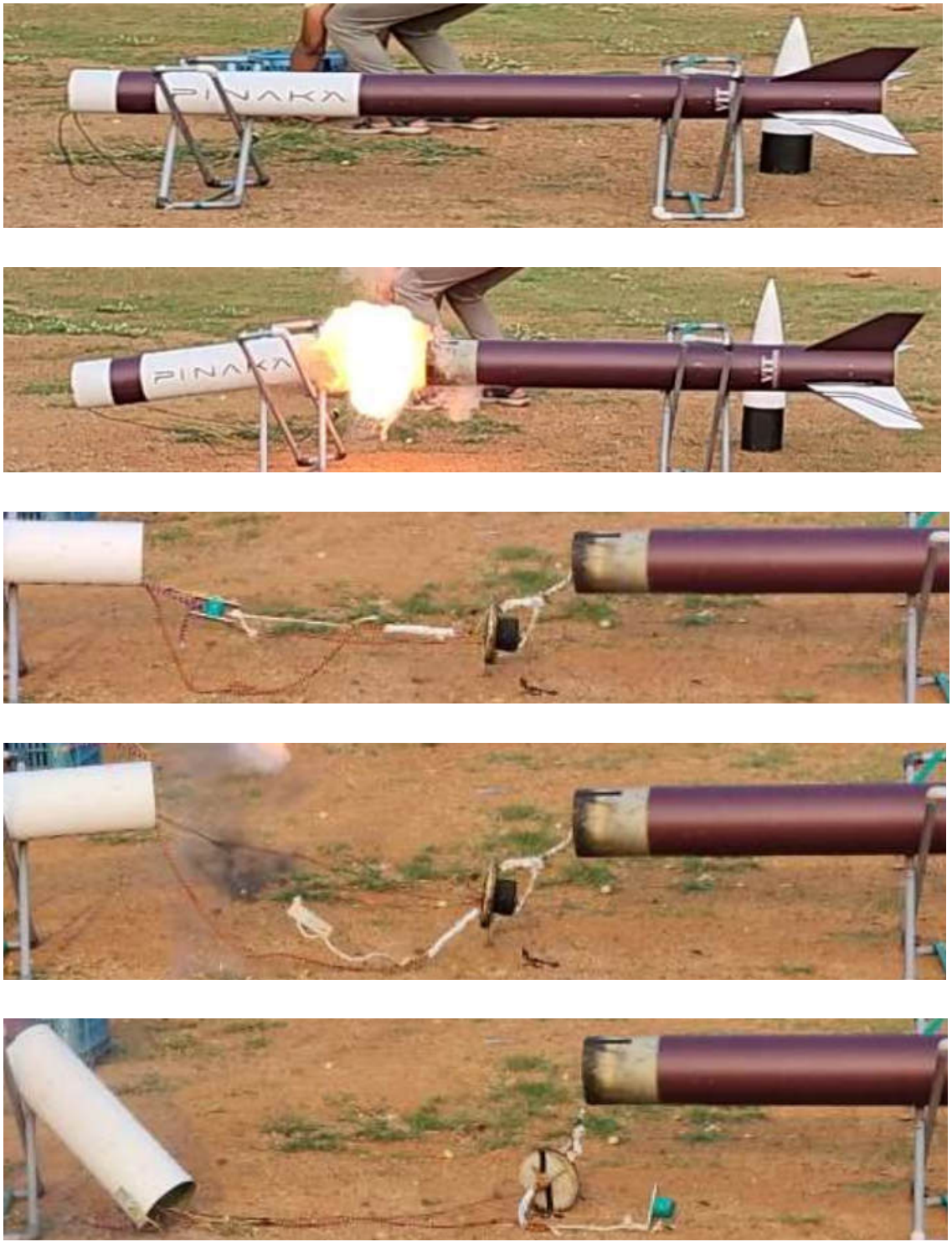


Fig: a) Before separation
b) During separation
c) After separation, before tender descender fires
d) During tender descender firing
e) Results

HAZARD ANALYSIS APPENDIX

Hazard	Effect/Severity	Plan of Action
General Safety Guidelines	**Mild to Severe**	Always ask a knowledgeable team member if you are unsure about the following: i. Equipment ii. Tools iii.Procedures iv. Handling of materials v. Other Questions you might have Be aware of your own actions and the actions of others i. Point out risks and mitigate them ii. Review procedures before beginning potentially dangerous actions. Safety Equipment i. Only wear hardened, sturdy shoes in the laboratory ii. Always wear safety glasses wherever possible iii. Always use respirators, i.e. face masks, respirators, etc., wherever possible iv. Always wear puncture-resistant gloves wherever possible e.g. when handling epoxy and other chemicals
Epoxy	Irritation of skin, eyes, and respiratory tract by contact and/or inhalation of hazardous vapors, sensitization of the skin. **Mild to moderate severity**	Use nitrile or latex gloves when handling epoxy, and only use it in a well ventilated area. If epoxy gets in your eyes, flush them immediately with water. Wash any skin that has encounter epoxy to avoid sensitization and irritation. If excessive fumes are inhaled, call the nearby hospital and take the patient to an ambulance.

Fiberglass	Irritation of the skin, eyes, nose, throat, respiratory tract, and digestive system from contact and/or inhalation of particles. **Mild to moderate severity.**	Use gloves and wear long sleeves when handling fiberglass. Cover mouth and nose with an examination mask or respirator and wear safety glasses when cutting. If fiberglass particles get in your eyes, flush with water immediately. Wash any skin that has come in contact with fiberglass particles to prevent irritation. If excessive particles are inhaled, call a nearby hospital and transport the patient to an ambulance.
Propellant	Burns, blunt force trauma from shock wave or engine case impact, lacerations from shrapnel. **Severe.**	Extreme care must be taken when handling the engine. Only trained personnel should handle and assemble the engine. The detonator will remain sealed and away from the engine until the vehicle is properly mounted on the rail and all personnel, except essential, have left the area. Ensure that the ignition circuit is open until all personnel have moved to the designated safety area. The engine will not be assembled until it is ready to be loaded into the body of the vehicle. Until then, it will remain in a cool, dry place in a sealed and locked container away from any flame or heat source. A first aid kit with bandages and burn treatment options must always be handy when handling the engine. 5 gallons of water and a fire extinguisher must be kept in reserve to extinguish fires and treat burns. All personnel should be trained in the treatment of burns and cuts. Should an emergency occur, immediately alert the nearby hospital while first aid is administered to the victim.

Dremel	Burns and lacerations. **Medium to severe.**	Avoid burns and lacerations by wearing long sleeves and gloves when operating a Dremel. Always wear eye protection. Keep a first aid kit ready to treat burns and lacerations. Flush any burns immediately with cool water. Depending on the level of the emergency, call a nearby hospital and transport the patient to an ambulance.
Black powder	Explosive. **Severe.**	Avoid damaging lungs by wearing appropriate face masks and respirators. Fire Extinguisher should be always available. Use of Proper gloves and lab coats to avoid contact with the skin. Depending on the level of the emergency, call a nearby hospital and transport the patient to an ambulance.

RISK ASSESSMENT APPENDIX

Hazard	Possible causes	Risk of mishap and rationale	Mitigation approach	Risk after mitigation
Uncontrolled thermal runaway from Solid Rocket Motor including possible flying debris.	- Fractures in the propellant grain - Nozzle clogging - Manufacturing defects of the engine case	**Low**: COTS component, verified by experienced professionals	- Visual inspection of the motor housing - Visual inspection of the nozzle - Visual inspection of the fuel grain - Only necessary crew during operation	**Low**
Uncontrolled Rocket Trajectory with possible influence on persons or devices	- Fin failure - Structural failure of the nose cone - Buckling of body tube	**Medium:** Student-built aerostructures with limited testing capabilities.	- Visual inspection of empennage for manufacturing defects that may affect aerodynamics - Visual inspection of body tube and nose cone - Compliance with safety clearances and barriers during the launch process	**Low to medium** (due to potentially catastrophic consequences).
Failure to deploy and unfold any of the outlined stages of the recovery system. Rocket experiencing terminal velocity descent	- Failure of electronics - Cutting of wires - Failure of power Storage System (battery) - Parachute ropes become jammed during ejection - Material failure of the parachute	**Medium**: Student built parachute with limited testing.	- Check the parachute for tears or disassembly of seams. - Check all COTS components for signs of fatigue - Inspect the parachute chamber and reserve parachute mount for possible obstructions - Implement all wiring and redundancy guidelines	**Low to medium** (due to the potentially catastrophic consequences).

Loss of the vehicle due to communication errors	- COTS avionics failure (software or hardware). - Severing of wires - Failure of the Energy Storage system	**Medium**: COTS avionics dictates transmission of data to the ground station	- Implement all cabling guidelines - Perform communication checks prior to launch	**Low**
Rocket does not ignite on command, but possibly on the approach of the crew for troubleshooting	- E-match has fallen out of position - Faulty e-match - Main power to the ignition system not present	**Low**: SRM and igniter are all COTS components	- Visually inspect the vehicle for signs of smoke or ignition - Ensure The main power to the ignition system is off - Restricted crew approach vehicle with appropriate PPE and barriers in place	**Low**
Vehicle loses mechanical support on the ramp, before takeoff	- Rail links failing structurally - Launch rail fails structurally	**Low**	- Inspect rail and fishplates before installing vehicle on-ramp - After vehicle installation, look for signs of rail or lug failure	**Low**

ASSEMBLY, PREFLIGHT, LAUNCH, RECOVERY, AND OFF-NOMINAL CHECKLISTS APPENDIX

A) Assembly checklist

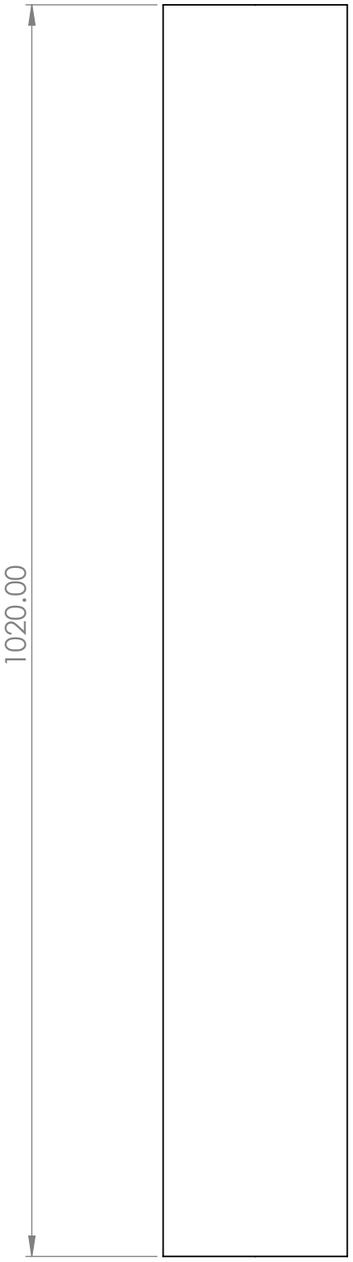
Step	Department	Task	Completion
1.0	**Recovery**	Fold and pack both parachutes properly.	☐
1.1	**Recovery**	Add black powder to recovery mechanisms.	☐
1.2	**Recovery**	Connect all shock cords according to diagram	☐
2.0	**Payload**	Assemble payload	☐
2.1	**Payload**	Attach payload to nose cone	☐
2.2	**Structures**	Screw in bulkheads for payload	☐
3.0	**Avionics**	Ensure all power connections are disconnected and flight systems are disarmed.	☐
3.1	**Avionics**	Confirm that all external wire connections are secure.	☐
3.2	**Avionics**	Install and secure the avionics in their designated sections within the body tube.	☐
3.3	**Avionics**	Securely mount the arming screws and establish necessary connections through the body tube.	☐
4.0	**Structures**	Attach bulkheads and eyebolts for recovery.	☐
4.1	**Structures**	Screw coupler with upper body tube.	☐
4.2	**Recovery**	Carefully place parachutes and connect shock cords.	☐
4.3	**Recovery**	Secure recovery and tender descender mechanism.	☐
4.4	**Structures**	Insert shear pins to connect lower body tube	☐
5.0	**Propulsion**	Obtain all required components for the motor.	☐
5.1	**Propulsion**	Assemble the motor.	☐
5.2	**Propulsion**	Insert assembled motor into the body tube.	☐
5.3	**Propulsion**	Attach the fin can/motor retainer.	☐

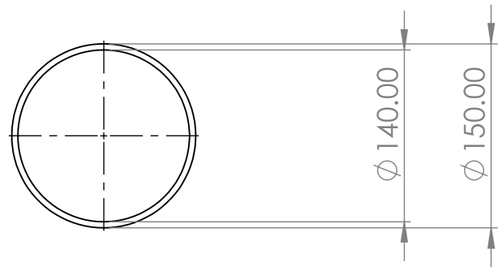
B) Pre-flight checklist

Step	Task	Check
1	Position the rocket on the launch pad.	☐
2	Activate the main power switch.	☐
3	Check for pre-arm status indications.	☐
4	Arm the avionics systems and verify status indications.	☐
5	Establish reliable telemetry communication with the Ground Station.	☐
6	Clear the launch pad for launch.	☐

C) Launch and Recovery

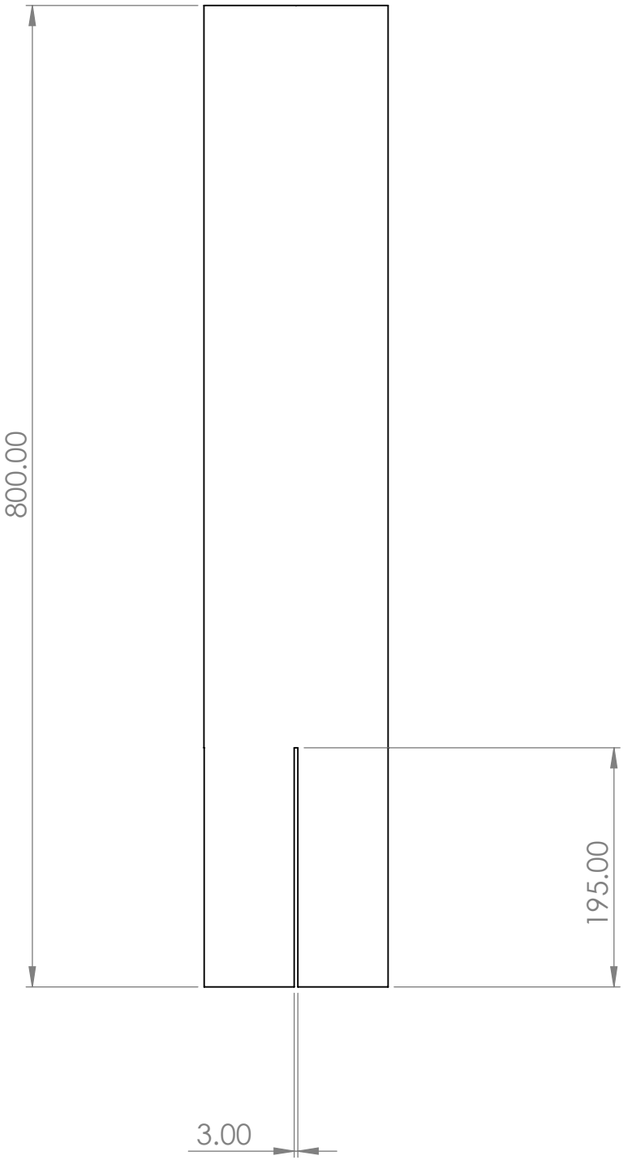
Step	Task	Check
1	Ignite the motor.	☐
2	Track rocket through telemetry and visual aid.	☐
3	Recover all sections of the rocket.	☐

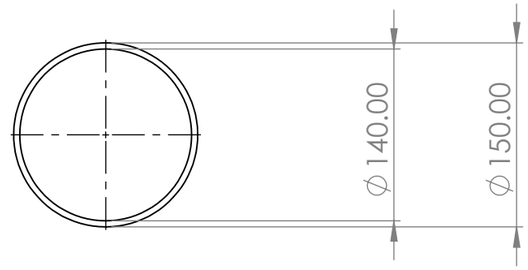




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CHK'D							
APPV'D							
MFG							
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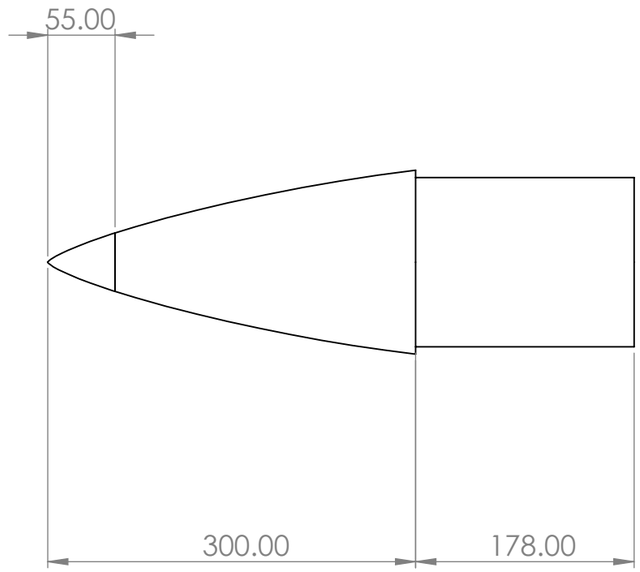
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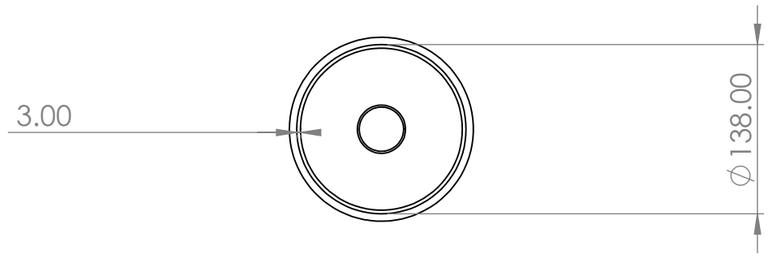


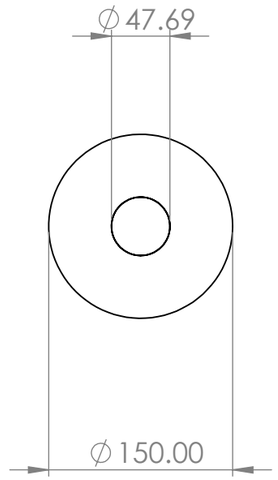


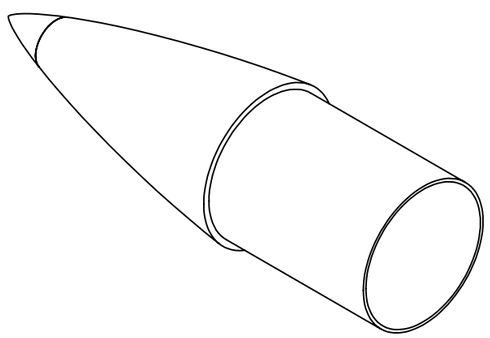
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CHK'D						
APPV'D						
MFG						
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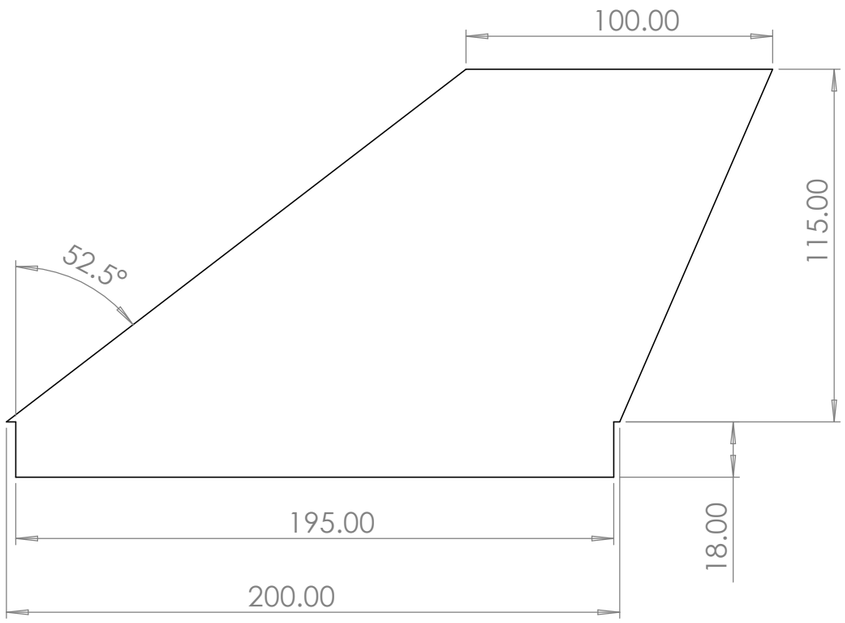


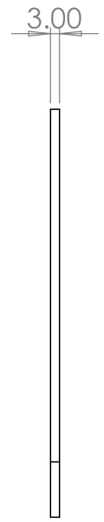


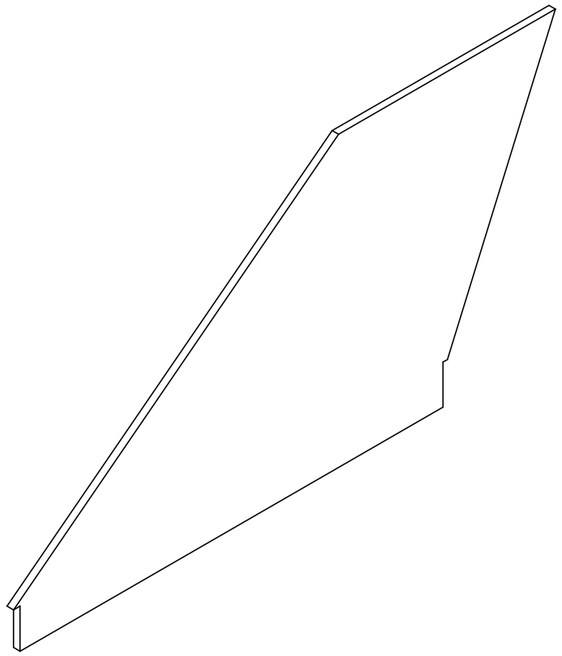


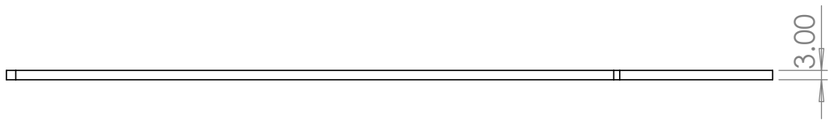
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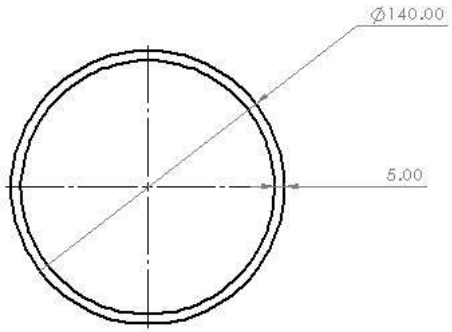




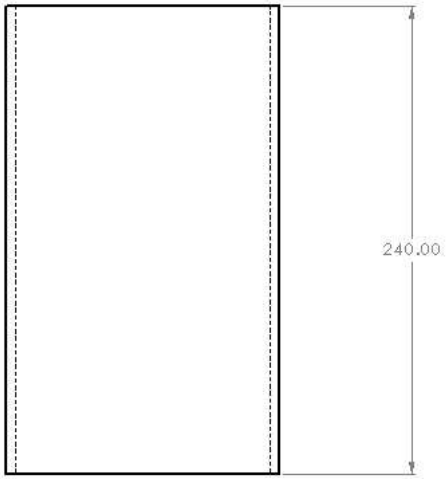


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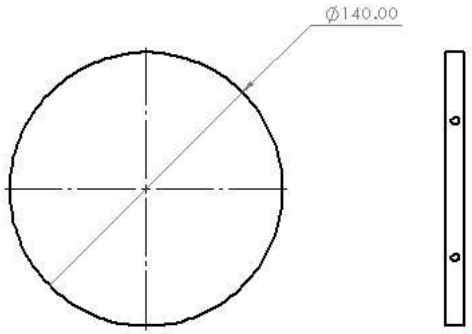




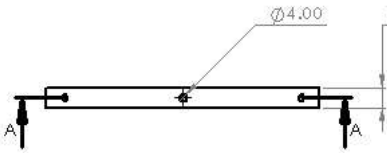


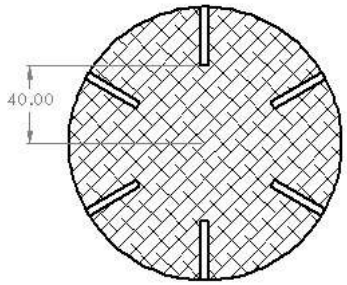
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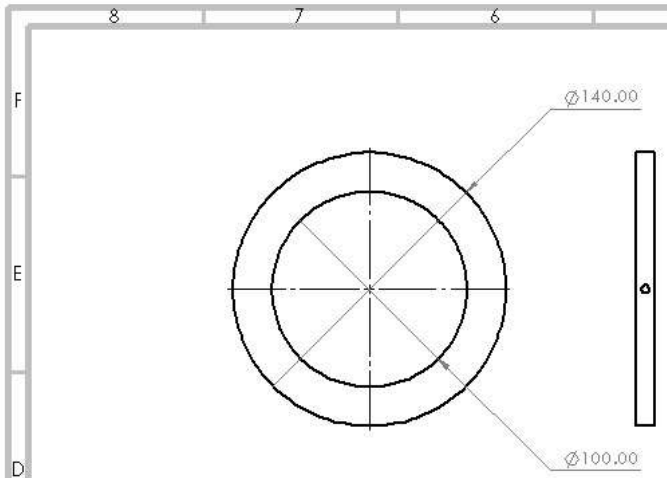


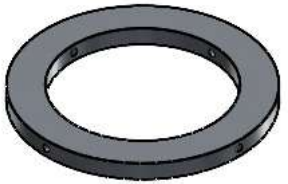
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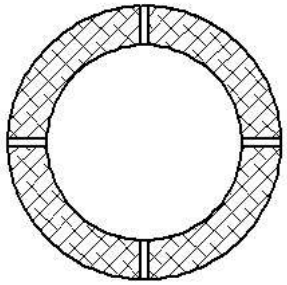
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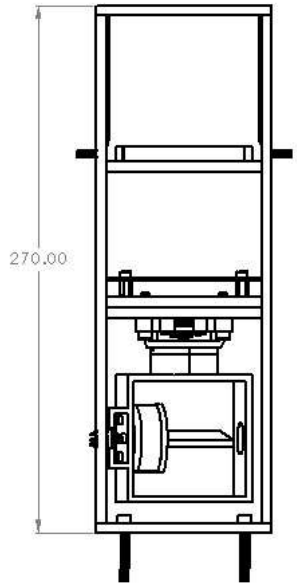


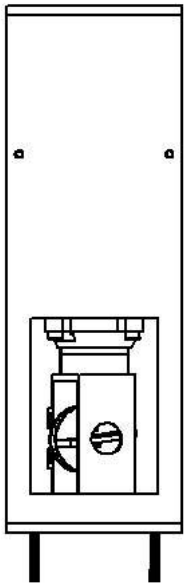
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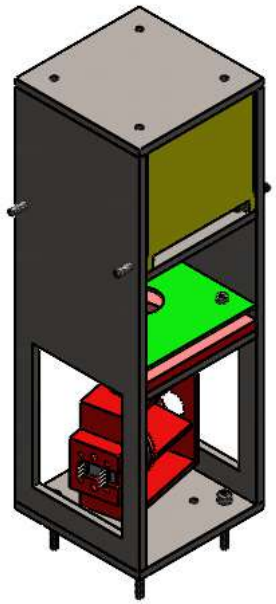
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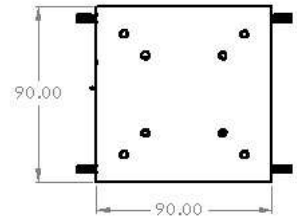
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CHK'D								
APPR'D								
MFG								
Q.A				MATERIAL:			DWG NO. Recovery Lip	A3
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LINEAR:
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FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

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CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

DWG NO

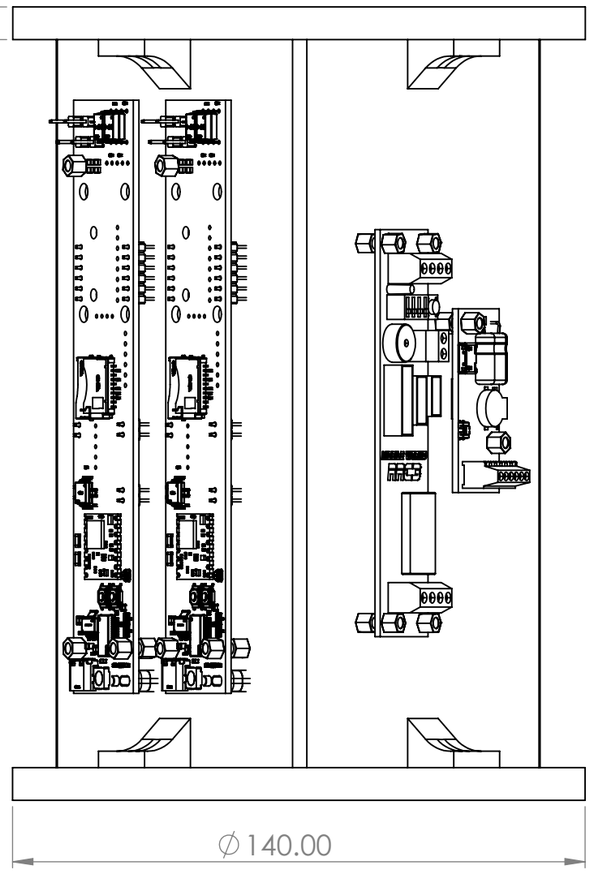
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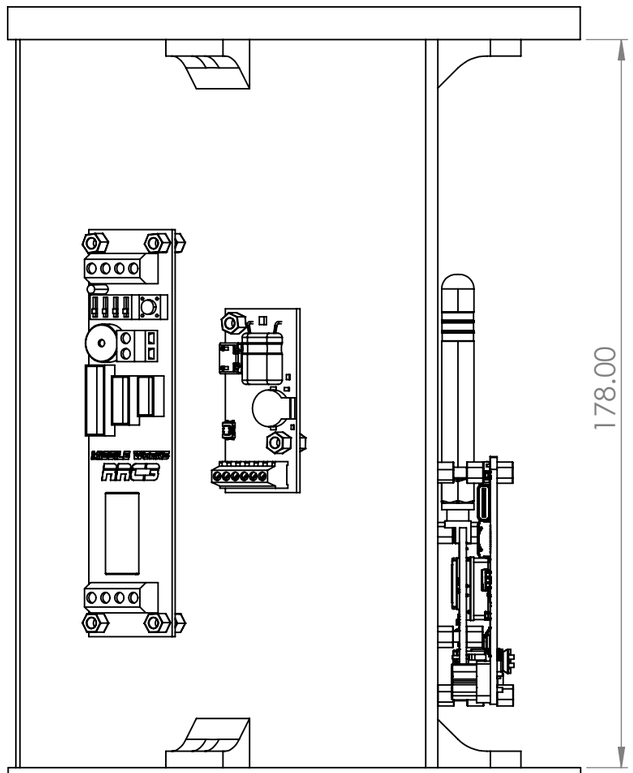
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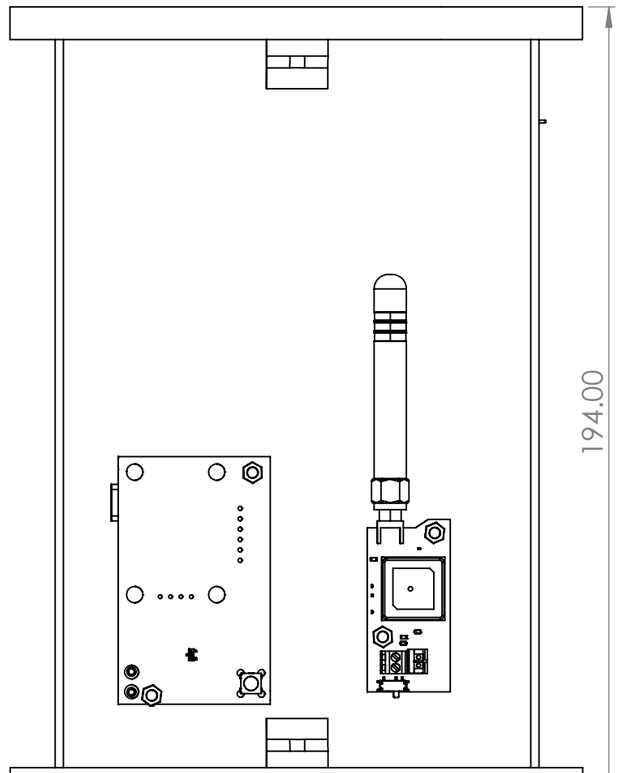
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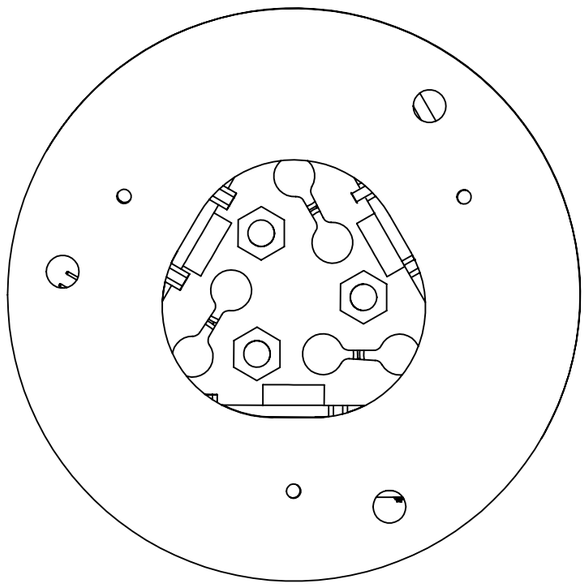
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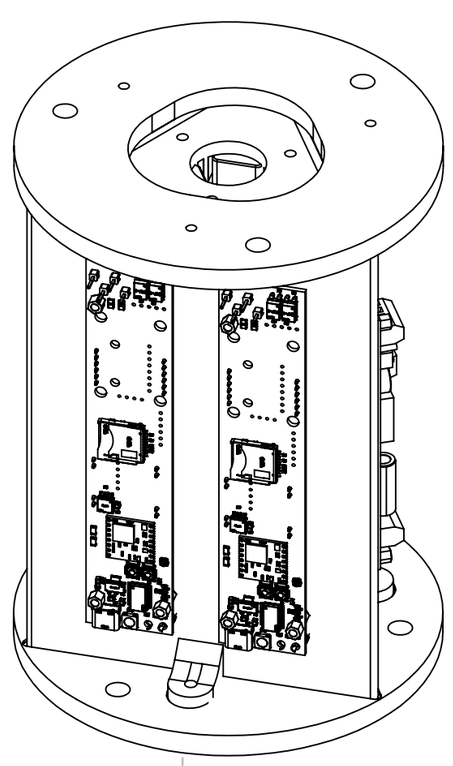
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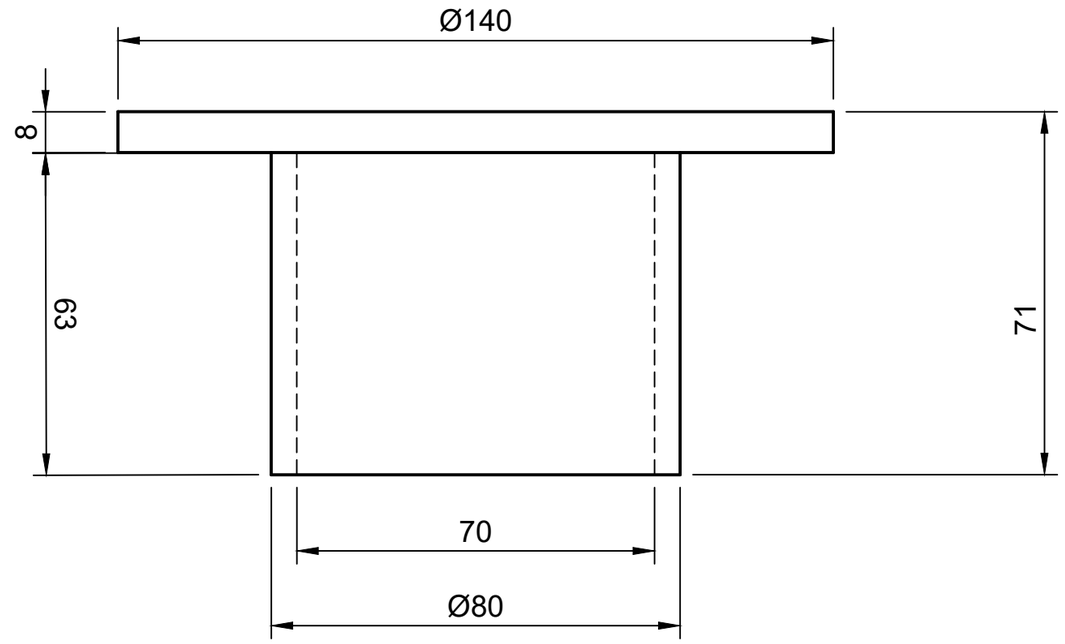


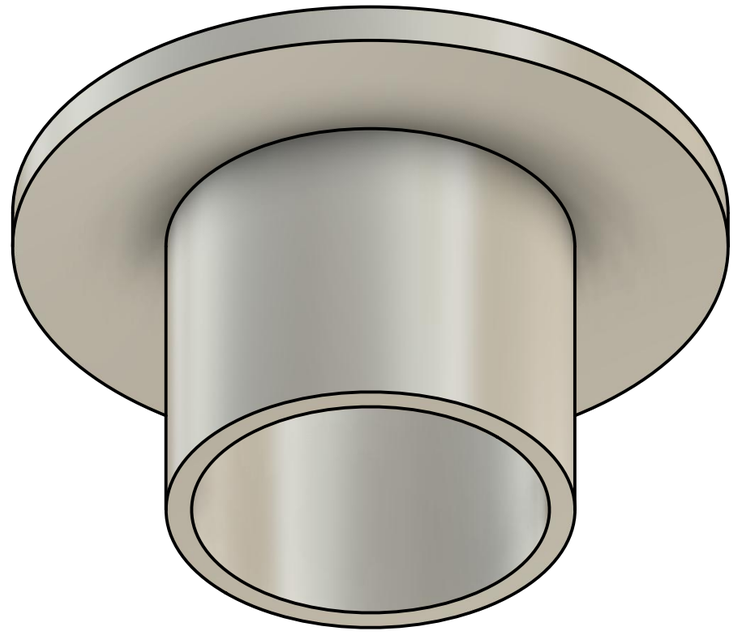


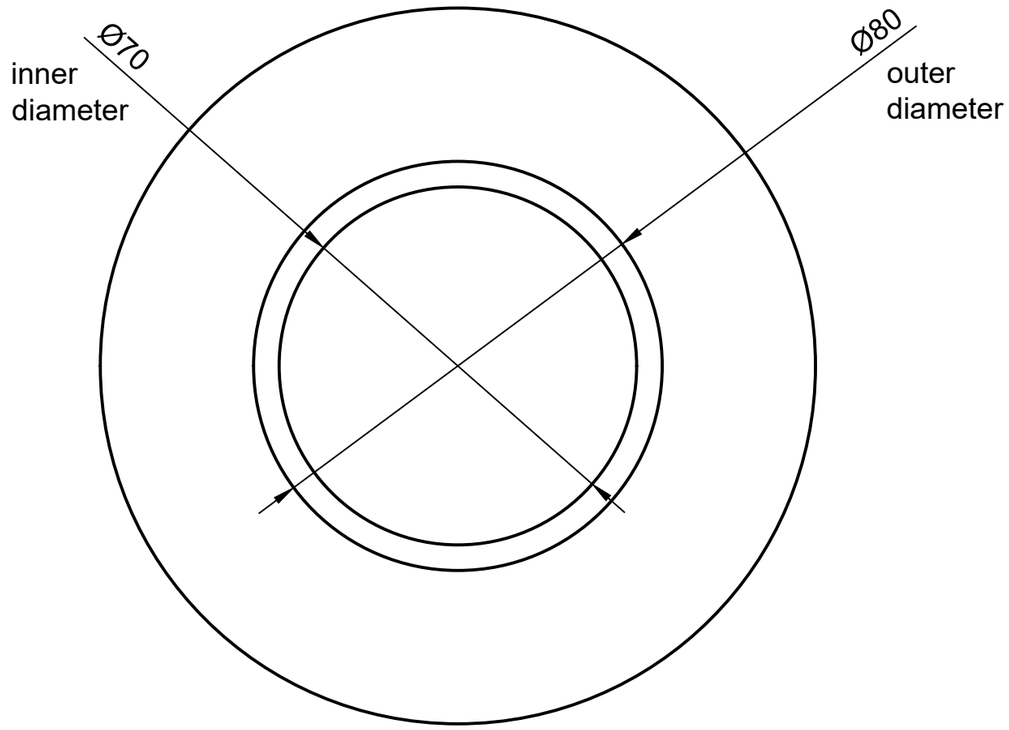


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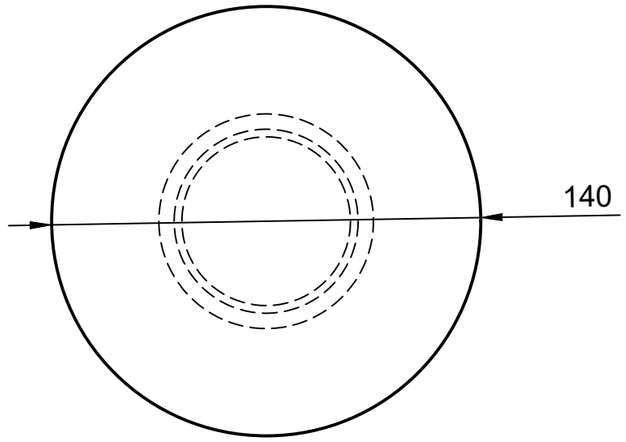
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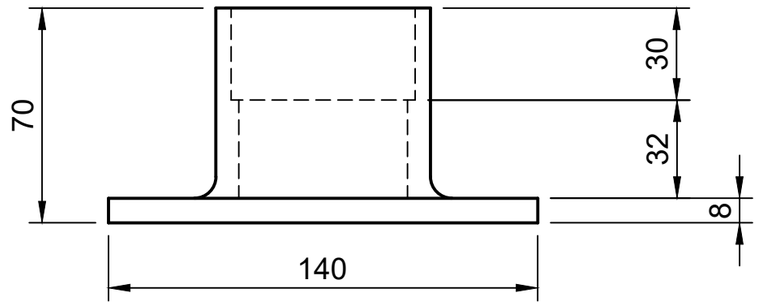


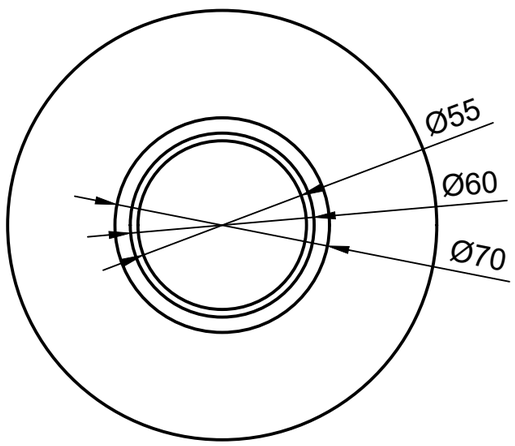


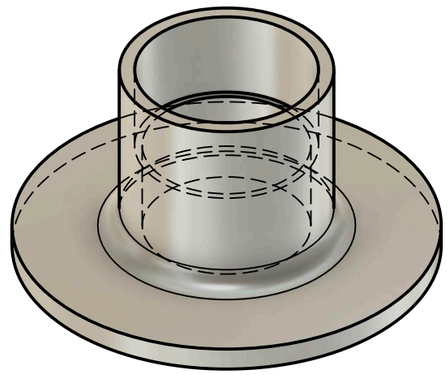


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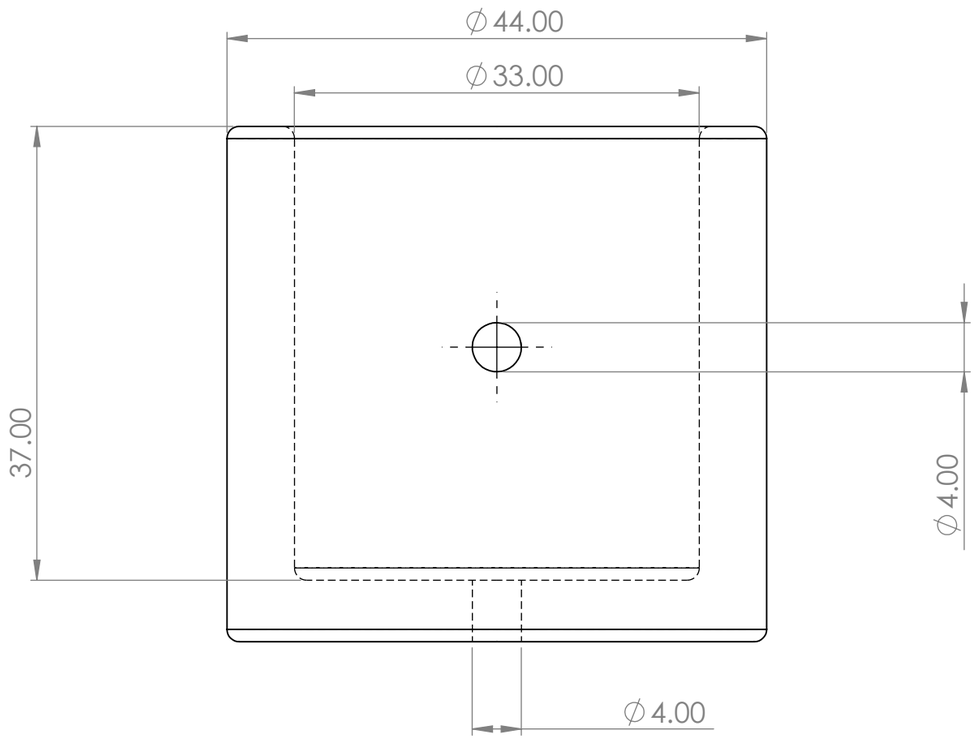


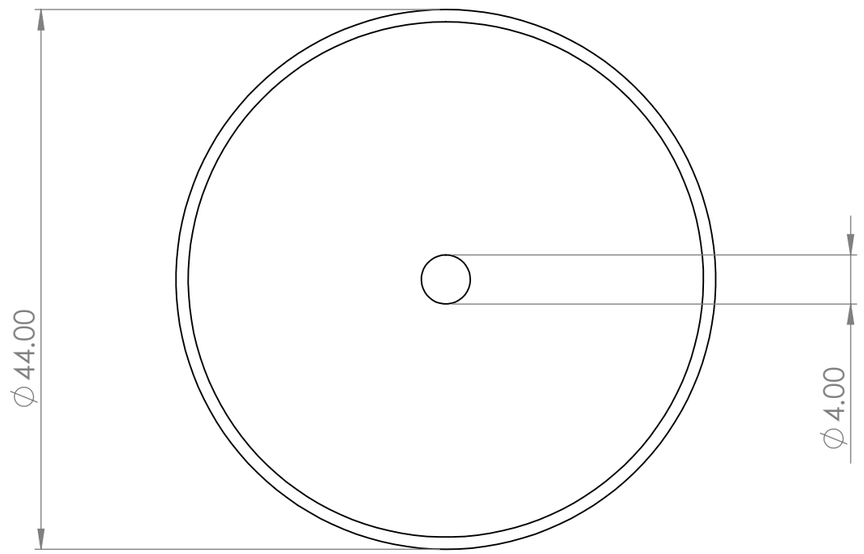




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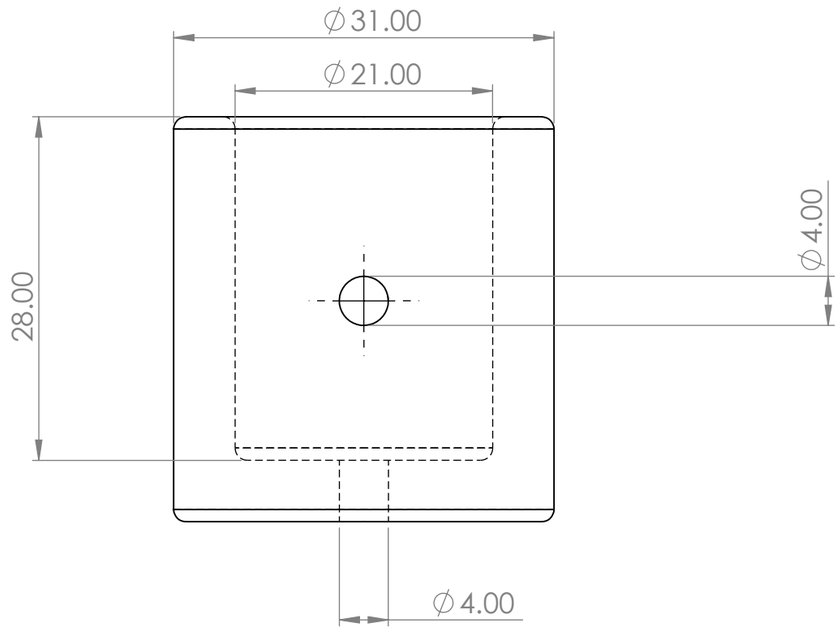
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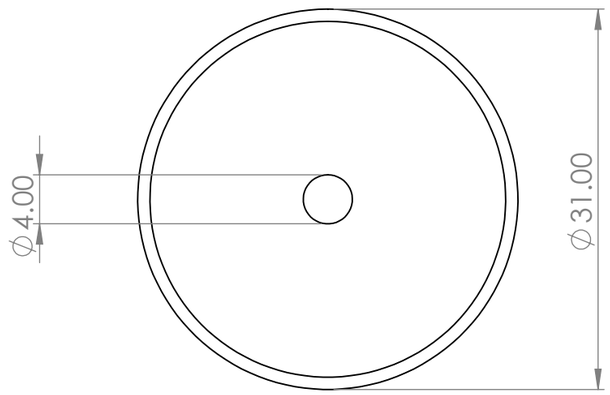




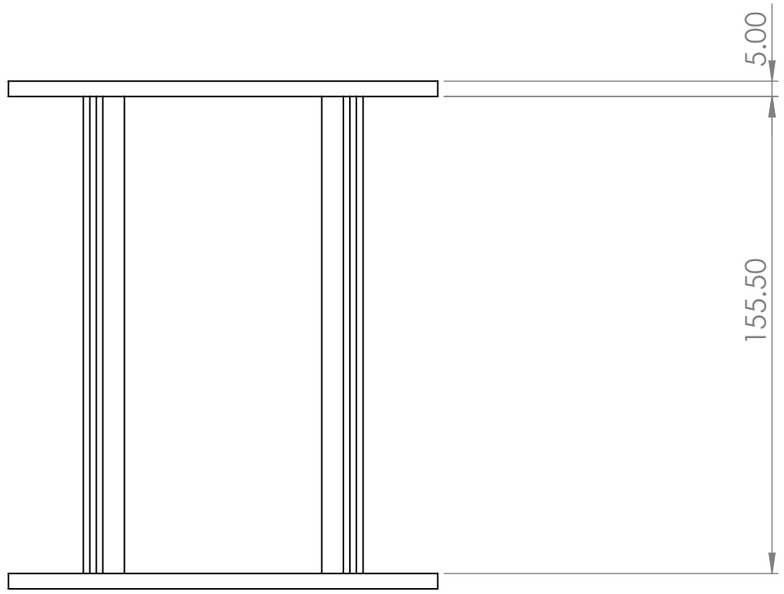
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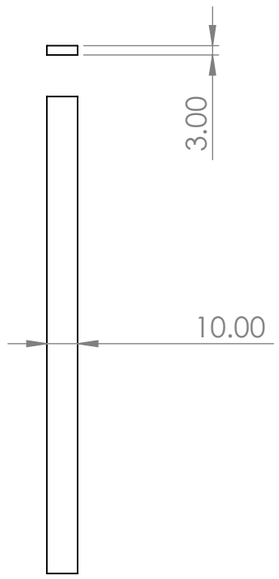
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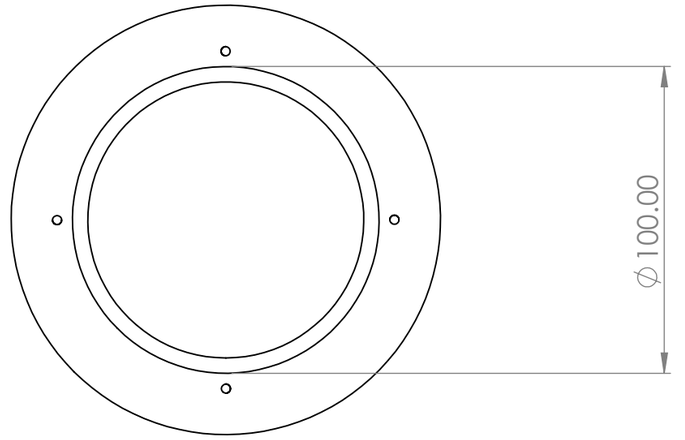


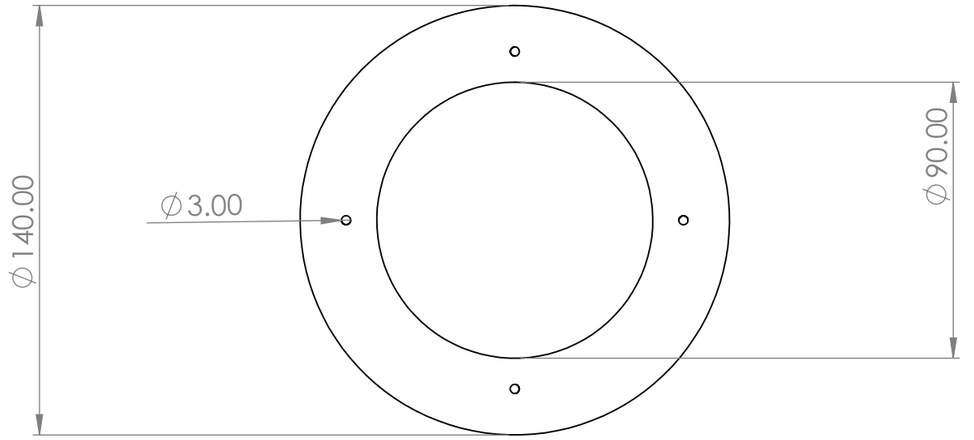


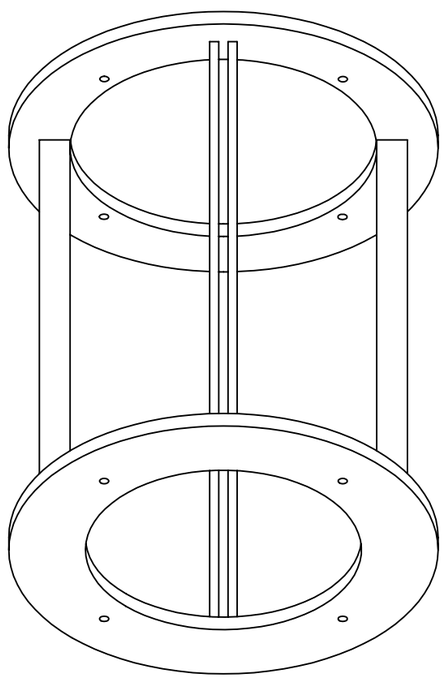
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